Spatiotemporal population assessment with open spatial capture-recapture and resource selection: A case study of rough-toothed dolphins

Philip T. Patton[1,2], Robin W. Baird[3], J. Andrew Royle[4], Erin M. Oleson[5], Kyleigh Fertitta[2], Tobias Friedrich[6], Michaela A. Kratofil[4], Sabre D. Mahaffy[4], Lisa C. McManus[2], and Lars Bejder[2]

[1] Migratory Bird Center, Smithsonian's National Zoo and Conservation Biology Institute, Washington, DC, USA
[2] Hawai'i Institute of Marine Biology, University of Hawai'i, Kāne'ohe, Hawai'i
[3] Cascadia Research Collective, Olympia, WA, USA
[4] Eastern Ecological Science Center, U.S. Geological Survey, Laurel, MD, USA
[5] Pacific Islands Fisheries Science Center, NOAA Fisheries, Honolulu, Hawai'i
[6] Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawai'i

Corresponding author: Philip T. Patton; PattonP@si.edu; Migratory Bird Center, Smithsonian's National Zoo and Conservation Biology Institute, PO Box 37012 MRC 5503, Washington, DC 20013-7012, USA.

Running headline: Spatiotemporal population assessment

Word count: 7025 out of 7000 words, 63 references.

Acknowledgments We would like to thank countless Cascadia Research Collective researchers for collecting and processing the photo-ID and satellite tag data for this manuscript. We would also like to express gratitude for our privilege to study these animals. The graduate assistantship for P.T.P. was funded by the NOAA Fisheries Quantitate Ecology and Socioeconomics Training (QUEST) Fellowship via the Cooperative Ecosystem Studies Unit (CESU) award NA19NMF4720181. All data were collected under strict animal welfare guidelines, including the US National Marine Fisheries Service permit XXXXXXXXXXXXXX and Institutional Animal Care and Use Committee YYYYYYYYYYYYY.

Authors' contributions **Philip T. Patton** Conceptualization, Methodology, Software, Formal Analysis, Visualization, Writing – Original Draft **Robin W. Baird** Data Curation, Funding Acquisition, Writing - Review & Editing **J. Andrew Royle** Conceptualization, Methodology, Validation, Supervision, Writing – Original Draft **Erin M. Oleson** Supervision, Funding Acquisition, Writing – Review & Editing, **Kyleigh Fertitta** Formal Analysis, Writing - Review & Editing **Tobias Fredrich** Formal Analysis, Data Curation, Writing - Review & Editing **Michaela A. Kratofi** Formal Analysis, Data Curation, Writing - Review & Editing **Sabre D. Mahaffy** Formal Analysis, Data Curation, Writing - Review & Editing **Lisa C. McManus** Supervision, Writing - Review & Editing **Lars Bejder** Conceptualization, Supervision, Funding Acquisition, Writing – Original Draft

Data Availability Statement The data and code necessary to rerun the analyses and generate the figures are available at ZZZZZZZZZZZ.

Conflicts of interest The authors declare no conflicts of interest.

Spatiotemporal population assessment with open spatial capture-recapture and resource selection: A case study of rough-toothed dolphins

Abstract

1. Spatial ecology plays an important role in population assessments. Populations are often defined by, among other things, the regions they occupy. This region can change over time if, say, the environment within the population's range changes, or individuals' relationships with their environment changes. This may be especially true for range restricted populations in highly dynamic environments, such as many marine mammal populations. Modeling this change is interesting from ecological and global change perspectives, and may be necessary to estimate unbiased trends in abundance for population assessments.

2. In this study, we conducted one such spatiotemporal assessment of the rough-toothed dolphin (*Steno bradenensis*) population that is associated with Kaua'i and Ni'ihau Islands, part of the Main Hawaiian Islands. To do so, we developed an open spatial capture-recapture model that includes resource selection (an open SCR-RSF model). To our knowledge, SCR-RSF models have only been developed for closed populations, and therefore have not been used in assessments to model changes in abundance over time. We demonstrate here that, while computationally demanding, these models can be fit in minutes with the NumPyro package.

3. There was evidence that the population declined from 1625 individuals (90% Credible Interval: 1339, 1947) to 1059 individuals (90% CI: 786, 1444) between 2011 and 2020, potentially because of low survival and recruitment.

4. Dolphins appeared to select areas with high zooplankton concentration and of intermediate depth. As a result, there was substantial overlap with the southern portion of the U.S. Navy's Pacific Missile Range Facility, which contained 14% of all use intensity for the population.

5. *Policy implications* We developed a modeling framework for explicitly integrating spatial ecology into population assessments. Assessments often include estimates of a population's bound-

ary and its trajectory, and this framework enables scientific agencies to jointly estimate both. We demonstrated the effectiveness of this approach by assessing the Kaua'i and Ni'ihau rough-toothed dolphin population. While this population is not officially recognized as a demographically independent population, we found evidence that it could be in decline, and that its core use areas overlap with the U.S. Navy's Pacific Missile Range Facility.

Keywords: data integration, Bayesian, cetacean, habitat selection

1 Introduction

Spatial ecology plays an important role in population assessments, which underpin conservation and recovery plans for many protected species (Kareiva et al., 1990; Carroll and Miquelle, 2006; Fletcher and Fortin, 2018). Regular population assessments are mandated by many national legislative acts and international agreements, such as the U.S. Endangered Species Act, the U.S. Marine Mammal Protection Act (MMPA), the EU Marine Strategy Framework Directive, the International Union for the Conservation of Nature's Red List, and the EU Birds Directive, among others. The primary goal of these assessments is to understand the population status or trajectory over time, thereby tracking the population's progress to recovery. These assessments, either directly or indirectly, also report on the population's spatial ecology. For example, the U.S. Marine Mammal Protection Act defines a population (technically, a stock) by the region it occupies. Specifically, a stock is, "a group of animals of the same species or similar taxa within a common spatial arrangement that interbreed when mature." As such, understanding space-use is a prerequisite to defining and assessing a population under the MMPA. Further, estimating the relationship between space-use and habitat covariates helps to understand habitat selection, which can be used to define critical habitat (McLoughlin et al., 2010). Designating critical habitat is an important process in several legal acts, including the U.S. Endangered Species Act and the EU Birds Directive.

Just as the size of a population can change over time, the space it occupies can also change (Brown et al., 1996). Individuals in the population can begin to use new areas, effectively expanding its range. Conversely, individuals can cease to use areas the population previously occupied. For example, the amount of preferred habitat within the population's range can decline, causing the population to con-

tract. Moreover, individuals' relationships with their habitat might change over time (McLoughlin et al., 2010). This may be especially true of range-restricted populations in highly dynamic environments, whereby the strength of resource selection depends on the availability of habitat. For example, prey availability can vary spatially and temporally in marine environments, such that predators must select the patches with the highest prey density relative to space, even if this density is low, say, relative to previous time periods (McLoughlin et al., 2010; Courbin et al., 2018; Benoit-Bird, 2024). Additionally, human use, which itself may vary spatially and temporally, can interfere with habitat selection (Bejder et al., 2006; Patchett et al., 2025). As such, models of spatiotemporal habitat selection in these environments may have to consider interaction effects been habitat and time.

Modeling the dynamic relationship between habitat and space-use helps estimate a population's trajectory. Spatial variability in space-use induces individual heterogeneity in encounter probability, i.e., the probability of encountering an individual during a survey (Royle et al., 2013b; Gardner et al., 2022). This heterogeneity, among other factors, can bias estimates of abundance in capture-recapture models, which are one of the primary methods for estimating abundance for population assessments, and potentially mask trends in abundance. Spatial capture-recapture (SCR) models that include resource selection (SCR-RSF) reduce such bias by explicitly modeling encounter probability as a function of habitat covariates (Royle et al., 2013b). To our knowledge, these models have only been developed for closed populations, and therefore have not been used in assessments to model changes in abundance over time.

Our primary objective for this study was to demonstrate the effectiveness of open SCR-RSF models for population assessments. To do so, we used a case study involving the rough-toothed dolphins (*Steno bredanensis*) that are resident to the islands of Kauaʻi and Niʻihau, within the Hawaiian Archipelago, where many other cetacean populations are resident to particular islands or island areas (Baird et al., 2008; Kratofil et al., 2023). Movement, genetic, and social network data suggest that there are two such populations of rough-toothed dolphins in the Main Hawaiian Islands: one associated with Hawaiʻi Island (also known as the Big Island), and one associated with Kauaʻi (Figure 1) (Baird et al., 2008; Albertson et al., 2017; Kratofil et al., 2023). Resident behavior might be advantageous to these populations because, while waters surrounding the Hawaiian Islands are nutrient poor, cyclonic eddies upwell nutrients near the islands, one part of a phenomenon known as the "island mass

effect" (Doty and Oguri, 1956; Rii et al., 2008; Nencioli et al., 2008; Feloy et al., 2024). This, in turn, brings zooplankton, which feed rough-toothed dolphin prey species, likely mesopelagic squid and fish (Shaff and Baird, 2021). As a result, rough-toothed dolphins might preferentially use plankton-rich habitats linked to such dynamic oceanographic conditions, and thus their movements associated with such habitats are likely to also vary across spatiotemporal scales (Benoit-Bird et al., 2013). The U.S. National Marine Fisheries Service, which is responsible for assessing marine mammal populations under the MMPA, only recognizes one stock of rough-toothed dolphins in Hawai'i (Carretta et al., 2024). Assessing the Kaua'i and Ni'ihau population independently, however, is beneficial because it is exposed to unique local threats. For example, this population overlaps with the U.S. Navy's Pacific Missile Range Facility (PMRF), where naval exercises involve the use of mid-frequency active sonar (Henderson et al., 2026). There has been some evidence for an effect of sonar exposure on fine-scale movements of tagged rough-toothed dolphins (Henderson et al., 2026), and more apparent behavioral responses to sonar have been documented in dense-beaked whales (*Mesoplodon densirostris*) (Jacobson et al., 2022). As such, understanding the population's size and trajectory, as well as its space-use, may help managers weigh the importance of these threats.

2 Materials and Methods

2.1 Case study

The primary goal of the case study was to estimate the population trajectory for the Kaua'i and Ni'ihau rough-toothed dolphin population from 2011 to 2020. To that end, we estimated the effect of landscape-level covariates on space-use over time within our open SCR-RSF framework. While these effects could be of interest *per se*, in this case, we were primarily interested in their ability to predict space-use. Additionally, we used the model of space-use to propose a boundary for the population and evaluated how that boundary varied over time.

Cascadia Research Collective (CRC) conducted unstructured surveys, on small boats (5.8m to 8.2m), off of Kaua'i Island from 2011 to 2020. See Baird et al. (2008, 2024) for a detailed description of survey methods. Surveys were unstructured in the sense that they did not follow random and predefined track lines. Instead, CRC designed the surveys to simultaneously maximize cetacean en-

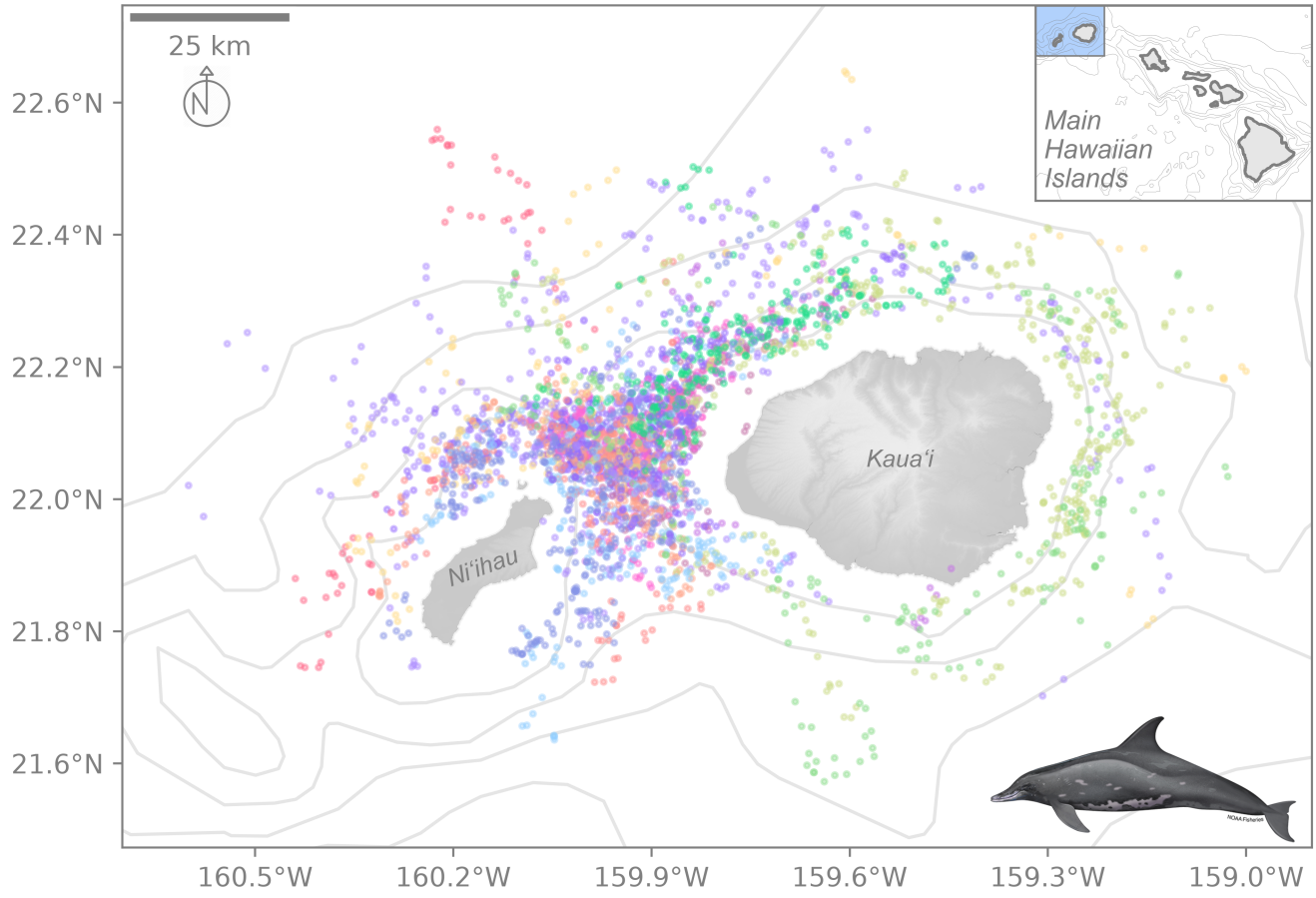


Figure 1: Argos-derived locations (colored dots) for 21 satellite tagged individuals from the Kaua'i Island population of rough-toothed dolphin. The color of the dot varies by individual. The depth contours (gray contour lines) represent 1000 m. Rough-toothed dolphin illustration courtesy of the U.S. National Marine Fisheries Service.

counters and spatial coverage. Most surveys took place off the western coast of Kaua'i (the lee), which is more likely to have suitable survey conditions (e.g., relatively low wind and swell) (Figure 2). On days where trade winds weakened or reversed, researchers were able to access and survey the windward side of the island. GPS locations of the survey vessel were noted every 5 minutes. Surveys occurred during dedicated field efforts, i.e., periods of one to three weeks where surveys were conducted near daily. This analysis included 14 field efforts and 147 survey days between February 2011 and February 2020 (Table B.1), primarily coinciding with PMRF naval activities.

Researchers approached every cetacean that they encountered to identify species, estimate group size, record locations, and collect photo-ID data (Baird et al., 2024). This rough-toothed dolphin population is partially marked, in that some animals can be identified using the distinctive nicks and notches along their dorsal fin and, as a secondary characteristic, pigmentation. Researchers took pho-

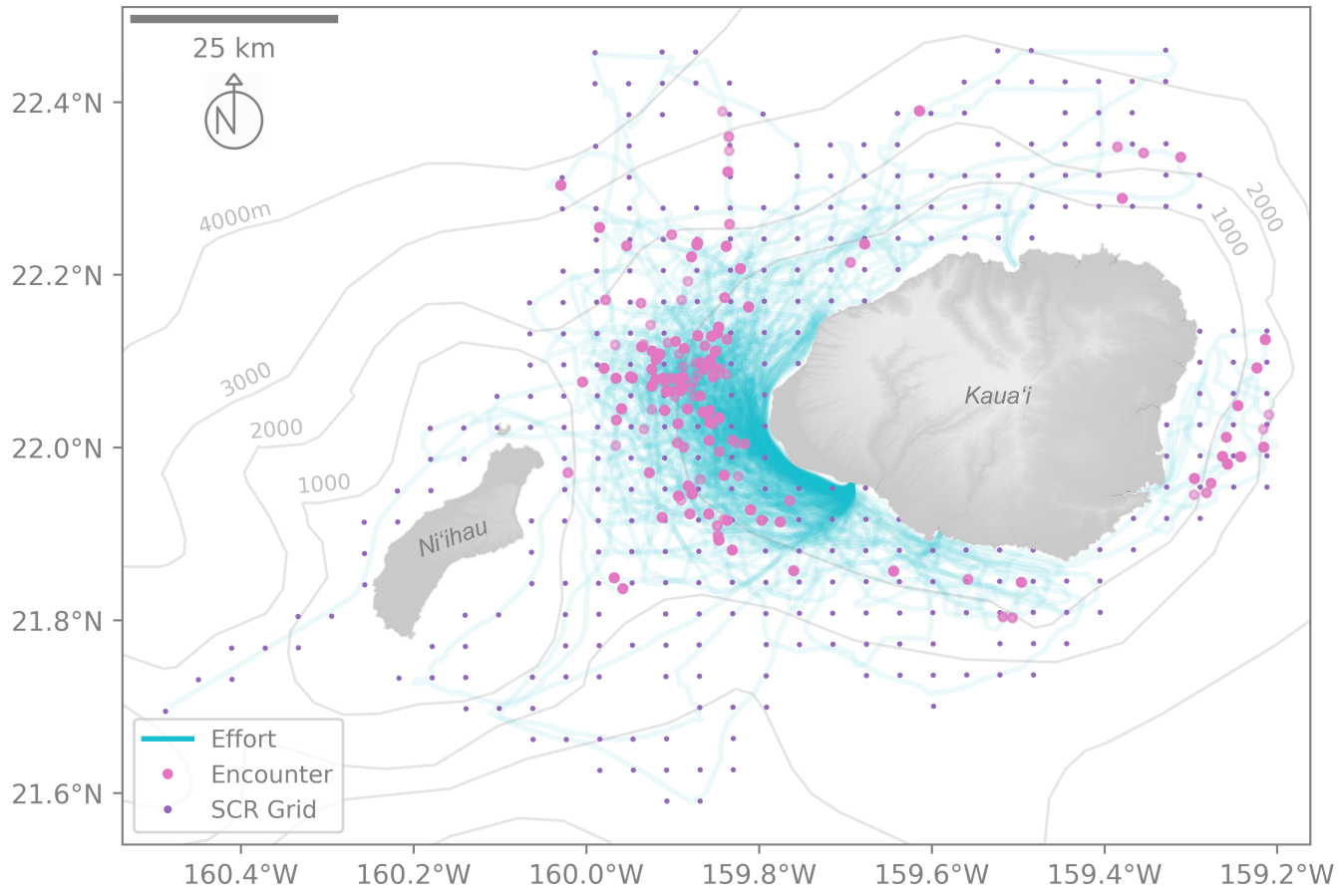


Figure 2: Map of survey effort (blue lines) and rough-toothed dolphin sightings (semi-transparent pink dots) near Kaua'i Island from 2011 to 2020. SCR requires continuous effort to be discretized onto a grid (purple dots). Each grid point is 4 km apart.

tographs of dolphins for the purpose of identifying them individually. CRC manually processed these images by classifying the image quality and the dolphin's distinctiveness. A detailed description of the photo-ID process can be found in Baird et al. (2008). We restricted this analysis to good or excellent quality images. We classified dolphins in such images as "marked" if CRC rated them as being distinctive or very distinctive (Rosel et al., 2011). The open SCR-RSF model is based on encounter histories of marked individuals. To estimate the proportion of marked individuals, i.e., the mark rate, researchers counted the number of marked individuals and the number of unmarked individuals in good or excellent quality images in each group of dolphins (see Section 2.5.1).

Researchers deployed 22 satellite tags on rough-toothed dolphins across the 14 field efforts, including 13 SPLASH10 tags, 6 SPOT5 tags, and 3 SPOT6 tags from Wildlife Computers (Figure 1). Location data were processed to remove erroneous locations following Kratofil et al. (2023). We thinned the telemetry data, regardless of location class, keeping one fix every six hours, to approximate re-

source selection decisions by each dolphin. To restrict the state space to just Kauaʻi, we excluded one telemetry deployment of a dolphin leaving Kauaʻi for Oʻahu, then returning to Kauaʻi. Following (Linden et al., 2018), we did not correct for errors in the telemetry fixes, but chose large enough pixels such that they would capture the telemetry error (see Section 2.2.1 and *Discussion*).

We investigated the effect of four covariates on space-use: bathymetric depth, log Apparent Fishing Effort (log AFE), vertical velocity, and zooplankton concentration (Figures C.1, C.2, and C.3). Each covariate was interpolated, when necessary, to a regular 4 km grid (see below). All covariates were standardized to have mean 0 and standard deviation of 1 (z-score standardization). There was no meaningful correlation between parameters (Figure C.4). Both zooplankton concentration and vertical velocity were allowed to vary by session. For these data, we constructed a session-level covariate out of daily data by averaging the covariate value for each date with effort—either days with photo-ID effort or days with telemetry fixes—in each session (Table B.1). For example, if a session had photo-ID effort on 02/07 and 02/08, and telemetry fixes on 02/07, 02/08, and 02/09, we averaged the zooplankton concentration across those three days. Unfortunately, the log AFE data was too sparse to match in this way; as such, we treated it as a static covariate across sessions. See Appendix A for a detailed description on generating the covariate data.

2.2 Open SCR model with resource selection

2.2.1 Open SCR model

The open SCR model is hierarchical, in that it can be separated into an observation model (encountering dolphins with photo-ID) and a process model (e.g., the births and deaths of individuals) (Gardner et al., 2010). We assumed that individuals entered the population and died according to a multi-state Jolly-Seber model (Jolly, 1965; Seber, 1965; Royle and Dorazio, 2008; Kéry and Schaub, 2011). In this parameterization, there are three existential states: not yet entered, $z_{i,t} = 0$; alive, $z_{i,t} = 1$; and dead, $z_{i,t} = 2$, for individual i during session t. In this case, each session corresponded to one of the primary periods, or years, in Table B.1, and we will refer to each session as "year." During each year, individuals transition from $z_{i,t} = 0$ to $z_{i,t+1} = 1$ with probability γ_t and die with probability $1 - \phi$ (Table 1). As

such, the existential state is categorically distributed,

$$z_{i,t+1} \sim \text{Categorical}\left(\mathbf{\Gamma}_{z_{i,t},\cdot}\right) \tag{1}$$

where $\mathbf{\Gamma}_{z_{i,t},\cdot}$ is the row of the transition probability matrix associated with the previous state (Table 1). The true number of individuals that were ever alive during the study, the superpopulation size, N, is unknown and, as a result, the upper limit of the index i is unknown. To deal with this, we assume that there is a hypothetical upper limit, M, for the superpopulation size, N, such that $i = 1, 2, \ldots, n, \ldots, N, \ldots, M$, where n is the number of individuals encountered. This is known as data augmentation (Royle et al., 2007). Data augmentation works by augmenting the encounter histories derived from the photo-ID data with $M - n$ all-zero encounter histories (see below). One way to justify data augmentation is to parameterize the model such that there is a discrete uniform prior for the superpopulation size, i.e., $N \sim \text{DiscreteUniform}\,(0, M)$. For example, we can parameterize γ_t in terms of ψ, the inclusion probability, and π_t, the entrance probability (Royle and Dorazio, 2008, Chapter 10). The inclusion probability represents the probability that an individual i ever enters the population, such that, $N \sim \text{Binomial}\left(M, \psi\right)$. The entrance probability dictates the number of entrants into the population at a given year, B_t, such that, $B_t \sim \text{Multinomial}\left(N, \{\pi_1, \pi_2, \ldots, \pi_T\}\right)$. For the first year, we calculate γ as $\gamma_0 = \psi\pi_0$. For all other years, we calculate γ as $\gamma_t = \psi\pi_t \,/\, \prod_{j=1}^{t-1}\left(1 - \gamma_j\right)$.

Table 1: Transition probability matrix, $\mathbf{\Gamma}$, for the multi-state Jolly-Seber model

		Not yet entered $z_{i,t+1} = 0$	Alive $z_{i,t+1} = 1$	Dead $z_{i,t+1} = 2$
Not yet entered	$z_{i,t} = 0$	$1 - \gamma_t$	γ_t	0
Alive	$z_{i,t} = 1$	0	ϕ	$1 - \phi$
Dead	$z_{i,t} = 2$	0	0	1

The second component of the process model relates to the distribution of dolphins in space. We assumed that each individual had an activity center, $\mathbf{s}_i$, which is the individual's average location over the period of study (Borchers and Efford, 2008; Royle and Young, 2008). $\mathbf{s}_i = (s_{i1}, s_{i2})$ is a two-dimensional spatial coordinate in easting (km) and northing (km). We assumed that each individual's activity center was constant across all years (Gardner et al., 2018). Moreover, we assumed that the activity centers were uniformly distributed across the state space, that is, $\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_M \sim \text{Uniform}\,(\mathcal{S})$.

The state space, $\mathcal{S}$, is an arbitrary polygon around the survey area, chosen to be sufficiently large that it does not bias $\hat{N}$ (Royle and Young, 2008). Figure 1 shows the state space that we used, which is a rectangle around extrema of the telemetry data. Constant and uniformly distributed activity centers seemed reasonable for a resident, island-associated population over the course of a year. This choice, compared to a moving activity center model, might bias our estimates of apparent survival (Gardner et al., 2018). Nevertheless, this may have been the best option given the population's ecology and the limitations of the study design (see *Discussion*).

For the observation model, we assumed that individuals can be encountered up to $K_{j,t}$ times at grid cell j during year t, where $K_{j,t}$ is the number of times that researchers searched grid cell j during year t, where $j = 1, 2, \ldots, 323$ (see below for the definition of "grid cell"). Allowing $K_{j,t}$ to vary in time and space accounts for the variable effort and the unstructured survey design (Efford et al., 2013). We assumed that the photo-ID encounter data followed a binomial distribution, with $y_{i,j,t} \sim \text{Binomial}\left(K_{j,t}, p_{i,j,t}\left[z_{i,t} = 1\right]\right)$, where $p_{i,j,t}$ is the probability that individual i is encountered in grid cell j during year t, and $[z_{i,t} = 1]$ is an indicator function that equals 1 when $z_{i,t} = 1$ and 0 otherwise. (Note that we represent the indicator function with the Iverson bracket $[\cdot]$ throughout.) We assumed that encounter probability declines with increasing distance to an individual's activity center, where the scale parameter, σ_t, determines the rate of decrease (Borchers and Efford, 2008; Royle and Young, 2008). Additionally, we assumed that dolphins preferentially selected habitats within their home range according to third-order resource selection and, as a result, were more likely to be encountered in such habitat (Royle et al., 2013b). For example, we hypothesized that dolphins would be more likely to be encountered in grid cells of intermediate depth, high vertical velocity, high log AFE, and high zooplankton concentration. We incorporated these hypothesized relationships in the linear model for encounter rate (Equation 4). All told, the SCR model can be expressed as,

$$y_{i,j,t} \sim \text{Binomial}\left(K_{j,t},\ p_{i,j,t}\left[z_{i,t} = 1\right]\right) \tag{2}$$

$$p_{i,j,t} = 1 - \exp\left(-\lambda_{i,j,t}\right) \tag{3}$$

$$\lambda_{i,j,t} = \exp\left(\alpha_{0t} - \alpha_{1t} d_{i,j}^2 + \alpha_2 \text{depth}_j + \alpha_3 \text{depth}_j^2 + \alpha_4 \text{fish}_j + \alpha_{5t} \text{velo}_{j,t} + \alpha_{6t} \text{zoo}_{j,t}\right) \tag{4}$$

where α_{0t} is the baseline encounter rate during year t, $\alpha_{1t} = 1/2\sigma_t^2$, $d_{i,j} = \|\mathbf{s}_i - \mathbf{r}_j\|$ is the Euclidean

distance between $\mathbf{s}_i$ and $\mathbf{r}_j$, $\mathbf{r}_j$ represents the two-dimensional spatial coordinates of grid cell j, depth_j is the bathymetric depth at grid cell j, fish_j is the log of the total log AFE at j, $\text{velo}_{j,t}$ is the vertical velocity at j during t, and $\text{zoo}_{j,t}$ is the zooplankton concentration at j during t. Equation 3 represents a Gaussian hazard encounter model, in that it implies a Gaussian (bivariate normal) model of space-use around $\mathbf{s}_i$ when $\alpha_2 = \alpha_3 = \alpha_4 = \alpha_5 = \alpha_6 = 0$.

In SCR models for unstructured surveys, each $\mathbf{r}_j$ represents a point in a regular grid across $\mathcal{S}$, where the spacing between each grid cell, $\Delta\mathbf{r}_j$, is a hyperparameter that must be set *a priori* (Russell et al., 2012; Milleret et al., 2018). Smaller $\Delta\mathbf{r}_j$, is more computationally expensive, while larger $\Delta\mathbf{r}_j$, may bias $\hat{N}$ (Milleret et al., 2018). We chose the largest possible $\Delta\mathbf{r}_j$, such that $\hat{N}$ was roughly the same for $\Delta\mathbf{r}_j + 0.5$ and $\Delta\mathbf{r}_j - 0.5$. Additionally, we ensured that $\Delta\mathbf{r}_j$ was smaller than $\hat{\sigma}$ (see below) (Milleret et al., 2018).

2.2.2 Resource selection function

We modeled the telemetry data using an RSF model consistent with equation 4. We discretized the state-space $\mathcal{S}$ into grid cells indexed by k. The centroid of each cell is located at $\mathbf{c}_k$, given in easting and northing coordinates (km). Let $x_{m,k}$ represent the total number of telemetry fixes in each grid cell k for each tagged individual m. We treated $x_{m,k}$ as a multinomial random variable, i.e., $x_{m,k} \sim \text{Multinomial}(R_m,\ \zeta_{m,k})$, where R_m is the total number of telemetry fixes from the tagged individual m, and $\zeta_{m,k}$ is the probability of the fix occurring within that cell. We modeled the use probability, $\zeta_{m,k}$, as a linear function of distance to the individual's home range center and habitat covariates,

$$x_{m,k,t} \sim \text{Multinomial}(R_m,\ \zeta_{m,k}), \tag{5}$$

$$\zeta_{m,k,t} = \frac{\exp\left(-\alpha_{1t} d_{m,k}^2 + \alpha_2 \text{depth}_k + \alpha_3 \text{depth}_k^2 + \alpha_4 \text{fish}_k + \alpha_{5t} \text{velo}_{k,t} + \alpha_{6t} \text{zoo}_{k,t}\right)}{\sum_{k=1}^{K} \exp\left(-\alpha_{1t} d_{m,k}^2 + \alpha_2 \text{depth}_k + \alpha_3 \text{depth}_k^2 + \alpha_4 \text{fish}_k + \alpha_{5t} \text{velo}_{k,t} + \alpha_{6t} \text{zoo}_{k,t}\right)}, \tag{6}$$

where $d_{m,k} = \|\mathbf{s}_m - \mathbf{c}_k\|$ is the Euclidean distance between each tagged individual's activity center $\mathbf{s}_m$ and $\mathbf{c}_k$. As such, this is a classical RSF model with a center of attraction, $\mathbf{s}_m$. Note that we indexed $x_{m,k,t}$ and $\zeta_{m,k,t}$ with t to highlight the fact that α_0, α_1, α_5, and α_6 can vary by year (see below); that said, no individual m was tagged during different years. The joint model consisting of 4 and 6 is an "integrated" model in the sense that it estimates the parameters affecting space-use with both the

photo-ID data and the telemetry data. In other words, the parameters α_2, α_3, α_4, α_5, α_6, and σ appear in both the RSF model and the open SCR model.

2.2.3 Candidate models

We considered 16 candidate models. Each model varied by whether it included time-varying or constant values for α_0, α_5, α_6, and σ (Table 2). We did not consider time-varying coefficients for the static covariates, depth and log AFE. Preliminary analysis showed no evidence of time variation in ϕ. We modeled time-variation in the α_0, α_5 and α_6 parameters as a random effect with a non-centered parameterization (Papaspiliopoulos et al., 2007) (see below for σ). For example, time variation in α_0 was modeled as

$$\alpha_{0t} = \mu_{\alpha_0} + \tau_{\alpha_0}\eta_{\alpha_0 t} \tag{7}$$

$$\eta_{\alpha_0 t} \sim \text{Normal}(0, 1) \tag{8}$$

$$\mu_{\alpha_0} \sim \text{Normal}(-5, 2) \tag{9}$$

$$\tau_{\alpha_0} \sim \text{HalfNormal}(1), \tag{10}$$

where each $\eta_{\alpha_0 t}$ represents the effect of each year, μ_{α_0} represents the average α_0 across years, and τ_{α_0} represents the variation across years. In this case, we expected the baseline detection probability to be quite low, hence the Normal$(-5, 2)$ prior. For α_5 and α_6, the model was the same except for the prior for the average, which we centered at 0, i.e., $\mu_{\alpha_5} \sim \text{Normal}(0, 1.75)$ and $\mu_{\alpha_6} \sim \text{Normal}(0, 1.75)$. We modeled time variation in α_1 as a fixed effect, such that $\log(\sigma_t) \sim \text{Normal}(3, 0.3)$, which placed 95% of the prior mass between 11.2 and 36.2 For reference, $\sigma = 36.2$km implies that the spends 95% of its time within about a 88km radius of it's activity center (Royle et al., 2013a, Chapter 5).

We used weakly informative priors for the other parameters to encourage sampling efficiency. For example, the priors for ϕ and ψ, Beta$(2, 2)$, encouraged the sampler to avoid the extreme boundary regions. We modeled each of the static covariate effects, α_2, α_3, and α_4 with Normal$(0, 1.75)$ priors. Finally, we modeled the entrance probabilities as $\boldsymbol{\pi} \sim \text{Dirichlet}(\mathbf{1})$, i.e., we assumed that individuals were equally likely to enter the population during any year. In reality, we expected most individuals to enter the population during the first year. These first year entrants are the individuals that were

Table 2: Candidate models, where t indicates that the parameter varied across years.

	α_0	σ	α_5	α_6
model-1	·	·	·	·
model-2	·	·	·	t
model-3	·	·	t	·
model-4	·	·	t	t
model-5	·	t	·	·
model-6	·	t	·	t
model-7	·	t	t	·
model-8	·	t	t	t
model-9	t	·	·	·
model-10	t	·	·	t
model-11	t	·	t	·
model-12	t	·	t	t
model-13	t	t	·	·
model-14	t	t	·	t
model-15	t	t	t	·
model-16	t	t	t	t

alive at the start of the study. We set $M = 6000$, which proved to be much larger than any MCMC sample from the posterior distribution of N.

2.3 Model checking and selection

We verified the fit of each model via posterior predictive checks (PPC). To that end, we used two test statistics that evaluated the spatial and temporal aspects of the model. We evaluated the statistics graphically and with a Bayesian P-value. In general, the P-value can be defined as $P = (1/V) \sum \left[d^{(v)} \leq d^{\text{obs}}\right]$, where $d^{(v)}$ is the v-th sample from the posterior predictive distribution of the test statistic, d^{obs} is the observed value of the test statistic, $\left[d^{(v)} \leq d^{\text{obs}}\right]$ is an indicator function that equals 1 when $d^{(v)} \leq d^{\text{obs}}$ and 0 otherwise, and $V = 6000$ is the number of posterior draws of $d^{(v)}$ (see below). The spatial statistic, d_{spatial}, was the number of telemetry fixes within the PMRF boundary. We sampled the spatial statistic with $d^{(v)}_{\text{spatial},t} = \sum_m \sum_{k'} x^{(v)}_{m,k',t}$ where $x^{(v)}_{m,k',t} \sim \text{Multinomial}(R_m, \pi^{(v)}_{m,k',t})$ is the predicted number of telemetry fixes for each in each grid cell, and k' is an index representing a grid cell within the PMRF. The temporal statistic, d_{temporal}, was the number of individuals encountered during each year. We sampled the temporal statistic with $d^{(v)}_{\text{temporal},t} = \sum_i \left[\sum_j y^{(v)}_{i,j,t} > 0\right]$, where we simulated $y^{(v)}_{i,j,t} \sim \text{Binomial}\left(K_{j,t},\ p^{(v)}_{i,j,t}\left[z^{(v)}_{i,t} = 1\right]\right)$. We defined failing the PPC as a model having a

$0.05 < P < 0.95$, which indicates poor fit. We suspected that even the best performing model might fail at least one of the PPC's because there were 18 test statistics (9 years of observed data, with a spatial and temporal statistic for each year). As such, we selected for inference the model (or models, in case of a tie) that passed the most checks.

2.4 Fitting models with NumPyro

We fit the open SCR-RSF model in a Bayesian framework using the Python package, NumPyro (Phan et al., 2019). NumPyro has two features that make it well-suited for hidden Markov models with many data points, such as the open SCR-RSF model. First, NumPyro can automatically enumerate, i.e., marginalize, discrete latent states like the existential state, $z_{i,t}$ (Obermeyer et al., 2019). Marginalized models tend to sample more efficiently than models with discrete latent states (Yackulic et al., 2020). Moreover, marginalized models can be sampled with the No-U-Turn Sampler (NUTS), which tends to be more efficient than, say, Metropolis-Hastings (Monnahan et al., 2017). Secondly, NumPyro relies on JAX, which compiles the model with just-in-time (JIT) compilation (Phan et al., 2019). In addition to making the gradient computation in NUTS more efficient, JIT accelerates repetitive likelihood calculations, such as $\|\mathbf{s}_i - \mathbf{r}_j\|$, through vectorized operations. JAX can execute these vectorized operations in parallel across every core on a GPU. A typical consumer GPU has between 4000 to 22000 cores.

We ran the No-U-Turn sampler for 500 warmup iterations, which we discarded, and 1500 post tuning iterations across 4 chains (6000 total posterior samples). We verified that the sampler converged with $\hat{R}$ (Vehtari et al., 2021), and monitored any divergent transitions during sampling (Betancourt, 2016). We ensured that the effective sample size was at least 200 per chain for all parameters, except the individual activity centers $\mathbf{s}_i$. We fit each chain with a NVIDIA Tesla V100 SXM2 GPU (5120 cores), which we accessed via Koa, the University of Hawai'i's High Performance Computing Cluster.

2.5 Inference and prediction

We used the model, or models (see below), of best fit to evaluate relationships between covariates and space-use, propose a population boundary, and estimate abundance over time. We defined abundance

as $N_t^{(v)} = \sum_{i=1}^{M} \left[z_{i,t}^{(v)} = 1 \right]$ for each posterior sample v, and $\hat{N}_t$ as the posterior mean of N_t. We defined the population boundary as the contour enclosing all grid cells with a relative use, u_k, of 0.5. Relative use is the amount of predicted use in a grid cell relative to a grid cell with average values for all covariates in equation 4 (i.e., average within the state-space) and the same distance to an individual's activity center. As such, an average grid cell has twice the use intensity of a grid cell with a $u_k = 0.5$. We predicted relative-use with the posterior samples of the α parameters, such that

$$u_{k,t}^{(v)} = \exp\left(\alpha_2^{(v)} \text{depth}_k + \alpha_3^{(v)} \text{depth}_k^2 + \alpha_4^{(v)} \text{fish}_k + \alpha_{5_t}^{(v)} \text{velo}_{k,t} + \alpha_{6_t}^{(v)} \text{zoo}_{k,t}\right), \tag{11}$$

where $u_{k,t}^{(v)}$ is the relative use of k at t for posterior sample v. We computed the population boundary with the posterior median, $\hat{u}_{k,t}$ for each session t, as well as an overall population boundary with the median use across all sessions and posterior samples, $\hat{u}_k$. To build a map of relative use with the population boundary, we "zoomed out" from the state-space by including all grid cells 100km from the center point between Kaua'i and Ni'ihau, purely for plotting purposes. We investigated the marginal effect of individual covariates on space-use by predicting the relative use as a function of the covariate and the posterior mean of the effect, holding all other covariates constant, e.g., $\hat{u} = \exp(\hat{a}_4 x)$ for values of $x \in (-2, 2)$. Recall that all covariates were standardized to have mean 0 and standard deviation of 1.

In the event that two or more models would perform equally well in the model checking procedure, we model averaged the predictions of abundance and of relative use. We assigned equal weight to all models. Thus, we computed the posterior median of the prediction, e.g., $\hat{N}_t$, across all posterior samples (i.e., if four models tied, we computed the median of the $4 \times 6000 = 24000$ posterior samples). We did the same for predictions of relative use when building the population boundary.

2.5.1 Estimating the mark rate and total population size

We predicted the total population size, i.e., the abundance marked and unmarked animals, N_t^{total}, with a Horvitz-Thompson-like estimator (Huggins, 1989; Badger et al., 2025). To do so, we computed, $N_t^{\text{total},(v)} = N_t^{(v)} / \theta_t^{(v)}$, where $\theta^{(v)}$ is the v-th sample from the posterior distribution of the "mark rate." The mark rate is the proportion of marked animals in the population. We estimated θ_j by modeling the number of marked individuals in good to excellent quality images, ω_l, in encounter l, as a binomial

random variable, such that $\omega_l \sim \text{Binomial}\left(\delta_l, p_l\right)$, where δ_l is the total number of individuals in good to excellent quality images in encounter l and p_l is the probability that an individual in the sample was marked. We expected the underlying mark rate to evolve over time, and therefore modeled it as a Gaussian Process. Additionally, we we included an encounter-level random effect, ε_l, because we expected over-dispersion. All told, the model was

$$p_l = \text{logit}^{-1}\left(\mu + f_{[\text{year}_l]} + \varepsilon_l\right) \tag{12}$$

$$f_t \sim GP(0, k) \tag{13}$$

where μ is the logit-scale grand-mean, ε_l is an encounter-level random effect, year_l indicates the year of sample l, and $k(t, t')$ represents the Matérn 5/2 covariance function. To get the mark rate for each year, we drew each sample of $\theta_l^{(v)}$ as $\theta_t^{(v)} = \text{logit}^{-1}\left(\mu + f_t\right)$. A detailed description can be found in Appendix D.

Finally, we investigated if there were evidence of a trend in abundance by computing the posterior distribution of the difference in abundance between 2020 and 2011. We did this for the marked population, i.e., $D = N_{2020} - N_{2011}$, and the total population, i.e., $D^{\text{total}} = N_{2020}^{\text{total}} - N_{2011}^{\text{total}}$. Additionally, we computed the posterior probability of a non-zero difference for both groups, e.g., $(1/V)\sum_v \left[N_{2011}^{(v)} \geq N_{2020}^{(v)}\right]$.

3 Results

CRC encountered 469 distinctive and very distinctive individual rough-toothed dolphins, in good or excellent quality images, from 2011 to 2020. Nine individuals were spatially captured, i.e., encountered at a different place at a different time, at least five times, 27 were spatially recaptured at least four times, and 282 were only captured once (Figure C.5). CRC visited $j = 1, 2, \ldots, 323$ total grid cells throughout the study period. The year with the most spatial coverage was 2015, when CRC visited 222 distinct grid cells (Figure C.6). The year with the most total effort was 2014, with 510 total cell/days of effort. The least spatial coverage, 51 grid cells, occurred in 2016, while the least total effort occurred in 2020, with 177 cell/days. CRC satellite tagged 18 dolphins (Figure C.7). The average tag deployment was 11 d and 15 h, with the longest deployment lasting 27 d 18 h and the shortest deployment lasting 3 d 18 h. After thinning, there were 738 telemetry fixes, i.e., $\sum R_m = 738$.

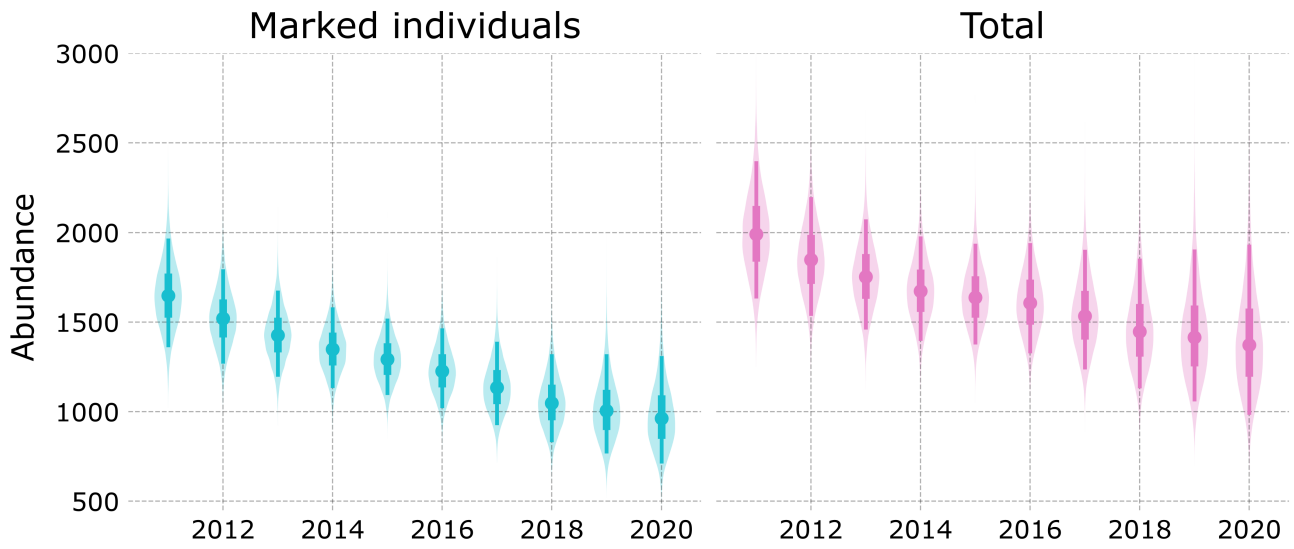


Figure 3: Posterior mean (dot), 50% credible interval (thick line), 90% credible interval (thin line), and posterior distribution (shaded area) of the number of marked individuals (blue) and the total number of individuals (pink) in the Kauaʻi, Niʻihau population of rough-toothed dolphins over time.

The number of marked individuals in the population appeared to steadily decline from 2011 to 2020, with $\hat{N}_{2011} = 1647$ (90% Credible Interval: 1360, 1967) and $\hat{N}_{2020} = 963$ (90% CI: 711, 1312) (Figure 3). The uncertainty in these estimates appeared to grow in the latter years, from 2011 ($\text{CV}(N)_{2011} = 0.11$) to 2019 ($\text{CV}(N)_{2019} = 0.17$) and 2020 ($\text{CV}(N)_{2020} = 0.19$), presumably because of the lack of sampling in 2019 and reduction in effort in the latter years (Tables 2 and C.6). Nevertheless, the number of marked individuals in the population appeared to remain the somewhat similar from 2018 to 2020, with $\hat{N}_{2018} = 1048$ (767, 1321). A more gentle decline was reflected in the estimate of the total population size, with $\hat{N}^{\text{total}}_{2011} = 1988$ (1629, 2396) and $\hat{N}^{\text{total}}_{2020} = 1373$ (990, 1941). This decline was mediated by the estimate of the mark rate, θ_t, which also declined during the study period, with $\hat{\theta}_{2011} = 0.82$ (0.78, 0.86) and $\hat{\theta}_{2011} = 0.69$ (0.59, 0.77) (Figure C.9). Even so, we estimated that the total population experienced a net loss of $\hat{D}^{\text{total}} = 607$ individuals (−50, 1190), with a posterior probability of decline of $P = 0.94$ (Figure 4). We estimated the net loss in the marked population to be $\hat{D} = 675$ individuals (233, 1098), with a posterior probability of decline of $P = 0.97$ (see *Discussion*).

The decline could be explained by low survival and recruitment. The estimate of apparent survival was 0.90 (0.87, 0.93). While low, this estimate should be interpreted cautiously (see *Discussion*). Nearly 75% of individuals were estimated to enter the population in the first year of sampling, $\hat{\pi}_{2011} = 0.72$ (0.62, 0.83), suggesting low recruitment. (Note that first year entrants include both re-

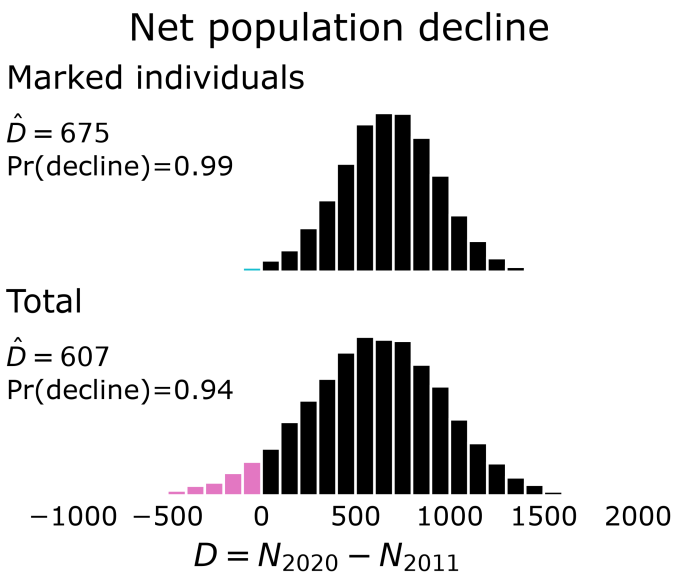


Figure 4: Posterior distribution (histograms) of the difference between the estimated population size in 2020 and 2011 for marked individuals (blue/back bars) and all individuals (pink/black bars). Bins containing samples less than zero are colored black. The posterior probability of decline (upper left text) is the proportion of negative samples.

cruits and existing members of the population.) Indeed, the estimates of entrants into the population, via recruitment or immigration, ranged from a low of $\hat{B}_{2018} = 27$ (2, 112) to a high of $\hat{B}_{2019} = 83$ (6, 310), although note that there was no sampling in 2019 (Figure C.10). Further note that these estimates were highly uncertain, with an average CV(B) of 0.93 from 2012 to 2020 (Table B.4).

We averaged the estimates of N and B across model-13 through model-16, which we deemed to be the models of best fit, despite the model selection procedure producing somewhat ambiguous results. Models with a constant α_0, i.e., model-1 through model-8, failed between 3 and 8 temporal PPCs (out of 9), suggesting that this was an important component of the detection process. Models with a constant σ and a time dependent α_0, i.e., model-9 through model-12, failed between 2 and 3 spatial PPCs (out of 9) while passing every temporal PPC. Model-13 through model-16 struck a balance, with each one failing one temporal PPC (2011) and one spatial PPC (2015). In each case, the model under-predicted the number of entrants in the first year, and under-predicted the number of fixes within the PMRF in 2015 (Figures C.11 and C.12). Including a stronger prior for the first occasion, e.g., $\boldsymbol{\pi} \sim$ Dirichlet ($\{4, 1, \ldots, 1\}$), may have improved model fit. It's unclear why the model predicted less use in the PMRF during 2015. To deal with this uncertainty, we performed inference and prediction by averaging results from model-13 through model-16.

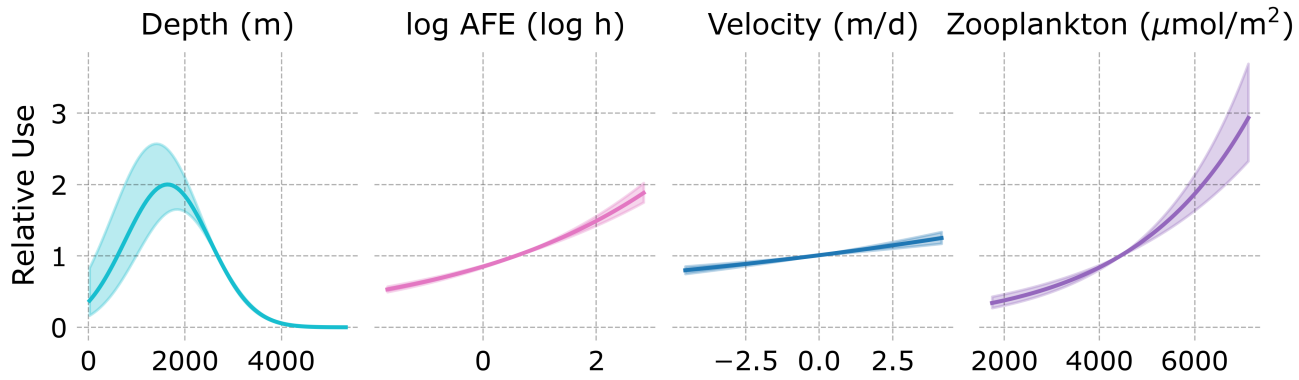


Figure 5: Posterior median of predicted relative use (colored line) and 90% credible interval (shaded area) at values of the four covariates. The covariate values span ±2 standard deviations away from the mean. A relative use of 1 indicates the grid cell has average use. Note, for example, that the average value for zooplankton concentration was 4435 μmol/m^2. "log AFE" refers to the log of Apparent Fishing Effort in a grid cell in hours. Velocity is the average vertical velocity in meters per day. Zooplankton is the concentration of zooplankton in μmol/m^2. Predictions assume average values for all other covariates.

On average, zooplankton concentration appeared to have the strongest effect on relative use (Figure 5). For example, the predicted relative use from model-13, assuming average values for all other parameters, was 2.9 (90% CI: 2.3, 3.7) at a concentration of 7128 μmol/m^2, which was 2 standard deviations above the mean concentration for the study period. Thus, dolphins were 3.1 times as likely to use a grid cell with 7128 μmol/m^2 than a grid cell with average values for all covariates (and the same distance to an individual's activity center; see Section 2.5 for a definition of relative use). Conversely, at 1742 μmol/m^2, 2 standard deviations below the mean, the relative use was 0.34 (90% CI: 0.27, 0.43). In model-16, the zooplankton effect varied from year-to-year, although was generally positive (Figure C.8). Similarly, in model-16, vertical velocity was largely positive and varied from year-to-year, although its effect was much smaller (Figure C.8). In model-13, the relative use at pixels with vertical velocity of 4 m/d was 1.3 (Figure 5). Depth and log AFE also appeared to correlate with space-use. Dolphins were most likely to be found at a depth of 1632 m, which had a relative use of 2.0 (90% CI: 1.6, 2.6). Space-use linearly increased with log AFE. For example, relative use was 1.9 (90% CI: 1.8, 2.0) in grid cells with high log AFE (20.1 h, or 3 log h).

The proposed population boundary, i.e., areas enclosed by the 0.5 relative use contour, contained the waters surrounding Kaua'i and Ni'ihau (Figure 6). The boundary enclosed 84% of the total use intensity for all grid cells in the map (Figure 6). The grid cells with the highest relative use were in the

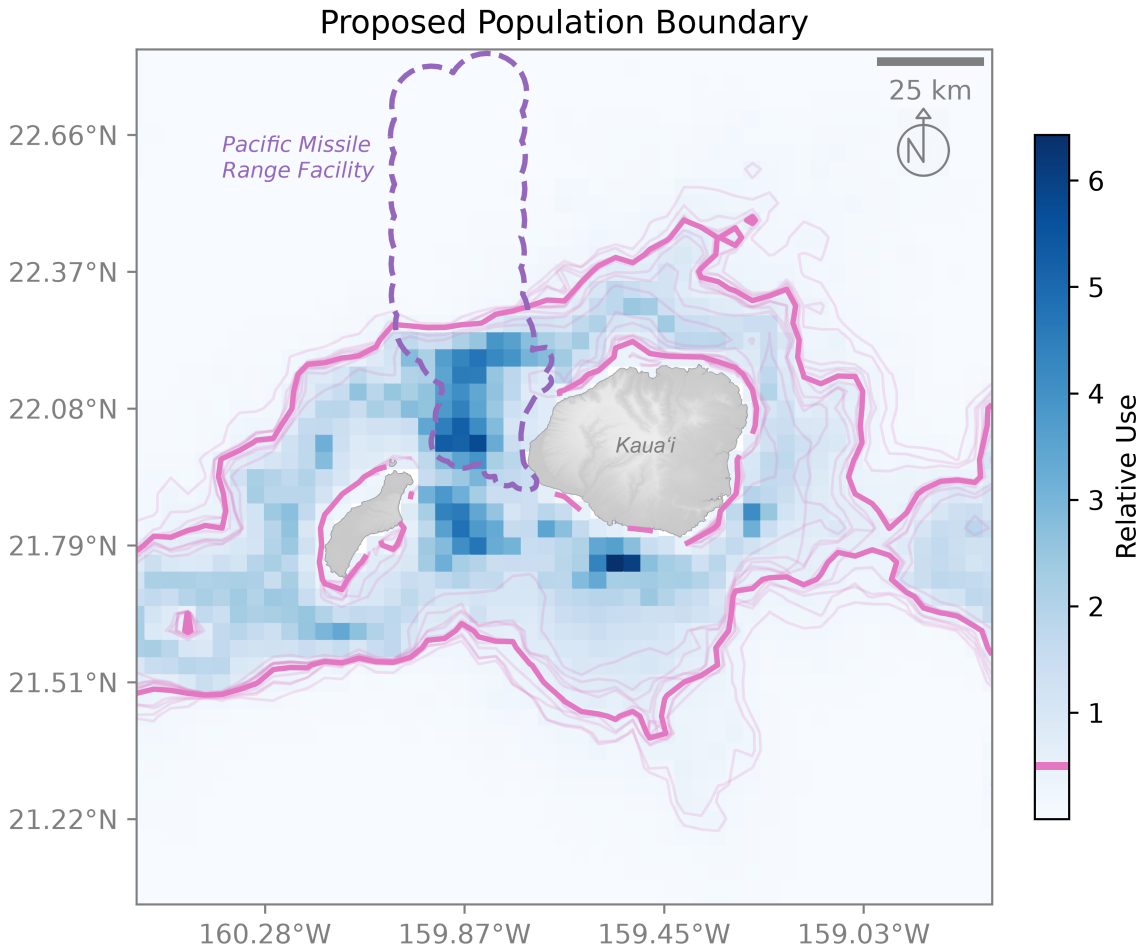


Figure 6: Posterior median of predicted relative use (blue cells), including a contour (thick pink line) enclosing grid cells with a relative use greater than 0.5, averaged across the entire study period. Year-to-year variation in use is depicted by the thin pink lines, which show the 0.5 contour for predicted relative use given the covariate values for each year. The border of the Pacific Missile Range Facility is shown as a purple dashed line.

Kaulakahi Channel that separates Kaua'i and Ni'ihau, as well as an area just south of Kaua'i. Some of the grid cells of highest intensity were within the PMRF, which included 14% of all use intensity. There was, however, a clear demarcation within the PMRF near the 3000m depth contour, where areas above this contour essentially had no use. There appeared to be some variation in the extent of the 0.5 use contour among years (Figure 6). The proposed population boundary appeared to grow and contract between years, presumably with variation in zooplankton concentration.

4 Discussion

We demonstrated the effectiveness of open SCR-RSF models at spatiotemporal population assessment through a case study of rough-toothed dolphins. Through the case study, we found that there was evidence that the population declined from 2011 to 2020, that population was most likely to use areas of high zooplankton concentration and intermediate depth, and Through the case study, we highlighted several steps, estimands, and predictions that might be of interest to users of this approach generally. For example, demonstrated how to develop spatiotemporal covariate structures for the model (Equations 6 and 8), select statistics for posterior predictive checks of the model (Figures C.11 and C.12), create a boundary for population's range (Figure 6), and how to fit the quickly with the NumPyro package (Phan et al., 2019). The case study showed how integrating photo-ID and telemetry data can help mitigate mismatches between a population's range and spatial effort (Figures 1 and 2), a common issue in population assessments of marine mammals (Badger et al., 2024), when estimating a population's trajectory (Figure 3).

4.1 Case study

We found evidence that the Kaua'i and Ni'ihau population of rough-toothed dolphins declined from 1988 individuals (90% CI: 1628, 2395) to 1372 individuals (90% CI: 989, 1940). Over the period, we estimated that the population experienced a net loss of $\hat{D}^{\text{total}} = 607$ individuals $(-50, 1190)$, with a posterior probability of decline of $P = 0.94$ (Figure 4). We observed the high probability of decline, despite the overlapping credible intervals for $\hat{N}^{\text{total}}_{2020}$ and $\hat{N}^{\text{total}}_{2011}$, because this metric essentially measures the percentage of posterior draws for which $\hat{N}^{\text{total}}_{2020}$ exceeded $\hat{N}^{\text{total}}_{2011}$ (Figure C.13). In terms of demographics, the decline appears to be the result of low recruitment and low apparent survival. Note, however, that this was not an estimate of true survival. The extreme spatial bias in photo-ID effort coupled with the short duration of the telemetry deployments prevented us from developing a realistic model of movement between primary periods. As such, this estimate, and hence the decline, could be explained by emigration from the study area. That said, that only one of the 22 telemetry deployments showed an animal emigrating, albeit temporarily, since it returned to Kaua'i during the deployment (Table B.2 and Figure C.14). Future research with greater spatial coverage of photo-ID or temporal coverage of telemetry would facilitate the estimation of a more realistic movement model

and reduce this uncertainty. We did not include field conditions into the detection model

Additionally, future research could explore potential causes of decline. We found that dolphins preferentially used areas with high zooplankton concentration. This overall concentration appeared to decline during the course of the study period, perhaps suggesting changes to the dolphins' base (Figure C.15). This evidence, however, is weak at best. Buttressing this evidence would require a more thorough investigation of changes of zooplankton concentration over the study period or, better yet, an investigation to the actual prey base of rough-toothed dolphins. That the prey base could be changing because of global change is conceivable. Marine heatwaves increase energetic demands of fish (Brijs et al., 2025), and have been implicated in declines of Hawaiian humpback whales (*Megaptera novaeangliae*) (Barber-Meyer et al., 2026). Additionally, future research could explore potential anthropogenic stressors including tourism, overlap with the naval base, and fisheries interactions. Although it's not obvious that any of these stressors would have changed in the study area from 2011 to 2020.

We found correlations between rough-toothed dolphin space-use and depth, log AFE, and zooplankton concentration. There may have also been a weak correlation with vertical velocity. Improvement in our estimates of zooplankton concentration and vertical velocity in the near-coastal areas may have improved estimates between these covariates and use (Figures 5, C.2, andC.3). The lack of data in near-coastal areas may have also hampered our ability to estimate between-year variation in habitat selection, hence the ambiguous model selection results (McLoughlin et al., 2010). Similarly, not accounting for the errors in the telemetry data likely biased our estimates of correlations with use towards zero (Brost et al., 2015). In any case, our map of the proposed population boundary showed notable overlap between the southern portion of the PMRF and a core-use area. Within the PMRF, evidence for behavioral responses to mid-frequency active sonar have been documented in some tagged rough-toothed dolphins (Henderson et al., 2026).

4.2 Spatiotemporal population assessment

Open SCR-RSF models can be valuable tools for population assessment because they can simultaneously estimate a population's abundance, demographic rates (e.g., apparent survival and recruitment), and its core use areas. Often, these are all requirements of environmental legislation, including the

MMPA. Core-use areas can also be estimated with descriptive methods, such as kernel density estimators, yet their lack of parametric effects and covariates limits their ability to, say, forecast use or extrapolate to nearby areas, and makes them susceptible to sample variance. Of course, forecasting with an open SCR-RSF model requires a rigorous approach to hypothesis and causal relationship development (Byrnes and Dee, 2025). While this was not an inferential objective of our case study, future users could easily adapt our approach to be more causal by, say, first developing a directed acyclic graph (DAG) of hypothesized drivers of space-use. This might obviate the need for non-causal covariates like log AFE, or at least clarify their relationship to estimands and predictors of interest.

Our case-study, along with recent developments in spatial partial-identity models, highlight some interesting questions for abundance estimation for cetacean populations via photo-ID. First, expanding this framework to continuous time would allow for the inclusion of field conditions into the detection model. Second, we restricted our analysis to good or excellent quality images of distinctive or very distinctive dolphins. This is standard practice because it minimizes the heterogeneity in detection probability induced by an individual's distinctiveness (Rosel et al., 2011). In other words, a highly distinctive individual is more likely to be recognized in a low quality image than an individual with less distinctive marks. So-called "random thinning SCR models" (Ferrão da Costa et al., 2026) assume that samples, e.g., photos, are "randomly thinned" such that they lose their identity information according to some probability, ρ, which in itself can be modeled as a function of covariates, e.g., a researcher's grade of individual distinctiveness or photo quality (Cove et al., 2023). Of course, this probability will be $\rho = 1$ for dolphins without any distinctive features. In this case, adapting traditional spatial mark-resight models in such a way that unmarked animals can become marked between sessions might be more fruitful (Royle et al., 2013a). Either way, incorporating these data in a model-based, data-generating fashion might improve estimates of the total population trajectory (Ferrão da Costa et al., 2026).

Finally, while SCR-RSF models are well-suited for cetacean studies—which often rely on unstructured survey designs and collect multiple types of data—they are equally applicable to any protected species where location tags are possible. In these cases, however, researchers might have to consider more realistic between year movement models. A simple option would be any of the circular kernels investigated by Efford and Schofield (2022), e.g., a bivariate normal random walk. Alterna-

tively, one could consider a settlement probability based on the habitat covariates in the RSF model (Bischof et al., 2020). A similar approach would be to apply the Langevin diffusion model of Gardner et al. (2022) to describe between-session movement rather than within-session movement. Both the Langevin and settlement approaches would describe second-order habitat selection, i.e., how individuals within the population choose home ranges within space, which we did not do here (Royle et al., 2013b).

New technologies, e.g., new location tags and the NumPyro package, have made open SCR-RSF analyses more accessible and efficient than ever. As such, there is an opportunity to improve the conservation and recovery of protected species by explicitly incorporating spatial ecology into population assessments. This may be especially important under global change because the relationship between habitat and space-use may be changing over time (McLoughlin et al., 2010).

References

Albertson, G. R., Baird, R. W., Oremus, M., Poole, M. M., Martien, K. K., and Baker, C. S. (2017). Staying close to home? Genetic differentiation of rough-toothed dolphins near oceanic islands in the central Pacific Ocean. *Conservation Genetics*, 18:33–51.

Badger, J., Baird, R., Johnson, D., Bradford, A., Mahaffy, S., Kratofil, M., Cullins, T., Currie, J., Stack, S., and Oleson, E. (2025). Accounting for sampling bias reveals a decline in abundance of endangered false killer whales in the main hawaiian islands. *Endangered Species Research*, 57:325–340.

Badger, J. J., Johnson, D. S., Baird, R. W., Bradford, A. L., Kratofil, M. A., Mahaffy, S. D., and Oleson, E. M. (2024). Incorporating telemetry information into capture-recapture analyses improves precision and accuracy of abundance estimates given spatiotemporally biased recapture effort. *Methods in Ecology and Evolution*, 15(10):1847–1858.

Baird, R. W., Mahaffy, S. D., Hancock-Hanser, B., Cullins, T., West, K. L., Kratofil, M. A., Barrios, D. M., Harnish, A. E., and Johnson, P. C. (2024). Long-term strategies for studying rare species: results and lessons from a multi-species study of odontocetes around the main Hawaiian Islands. *Pacific Conservation Biology*, 30(1).

Baird, R. W., Webster, D. L., Mahaffy, S. D., McSweeney, D. J., Schorr, G. S., and Ligon, A. D. (2008). Site fidelity and association patterns in a deep-water dolphin: Rough-toothed dolphins (*Steno bredanensis*) in the Hawaiian Archipelago. *Marine Mammal Science*, 24(3):535–553.

Barber-Meyer, S. M., Cheeseman, T., Barlow, J., Bejder, L., Bradford, A. L., Cartwright, R., Deakos, M., Evans, L., Finn, R., Gabriele, C. M., Goodwin, B., Hayslip, C., Henderson, E. E., Jones, M., Lammers, M., Lyman, E., McSwain, D. M., Mobley, J., Olio, M., Pack, A. A., Palacios, D. M., Spears, H. R., Stack, S. H., Sullivan, F. A., van Aswegen, M., Yano, K. M., Yin, S., Zoidis, A., and Currie, J. J. (2026). Humpback highs and lows amid shifting seas and a northeast pacific marine heatwave: Whale demographics in the main hawaiian islands, 2002–2023. *Marine Mammal Science*, 42(4):e70231. e70231 2731846.

Bejder, L., Samuels, A., Whitehead, H., Gales, N., Mann, J., Connor, R., Heithaus, M., Watson-Capps, J., Flaherty, C., and Krützen, M. (2006). Decline in relative abundance of bottlenose dolphins exposed to long-term disturbance. *Conservation Biology*, 20(6):1791–1798.

Benoit-Bird, K. J. (2024). Resource patchiness as a resolution to the food paradox in the sea. *The American Naturalist*, 203(1):1–13.

Benoit-Bird, K. J., Battaile, B. C., Heppell, S. A., Hoover, B., Irons, D., Jones, N., Kuletz, K. J., Nordstrom, C. A., Paredes, R., Suryan, R. M., et al. (2013). Prey patch patterns predict habitat use by top marine predators with diverse foraging strategies. *PloS one*, 8(1):e53348.

Betancourt, M. (2016). Diagnosing suboptimal cotangent disintegrations in Hamiltonian Monte Carlo. *arXiv preprint arXiv:1604.00695*.

Bischof, R., Milleret, C., Dupont, P., Chipperfield, J., Tourani, M., Ordiz, A., De Valpine, P., Turek, D., Royle, J. A., Gimenez, O., et al. (2020). Estimating and forecasting spatial population dynamics of apex predators using transnational genetic monitoring. *Proceedings of the National Academy of Sciences*, 117(48):30531–30538.

Borchers, D. L. and Efford, M. G. (2008). Spatially explicit maximum likelihood methods for capture-recapture studies. *Biometrics*, 64(2):377–385.

Brijs, J., Moore, C., Schakmann, M., Souza, T., Grellman, K., Tran, L. L., Patton, P. T., and Johansen, J. L. (2025). Eat more, often: The capacity of piscivores to meet increased energy demands in warming oceans. *Science of the Total Environment*, 973:179105.

Brost, B. M., Hooten, M. B., Hanks, E. M., and Small, R. J. (2015). Animal movement constraints improve resource selection inference in the presence of telemetry error. *Ecology*, 96(10):2590–2597.

Brown, J. H., Stevens, G. C., and Kaufman, D. M. (1996). The geographic range: size, shape, boundaries, and internal structure. *Annual review of ecology and systematics*, 27(1):597–623.

Byrnes, J. E. and Dee, L. E. (2025). Causal inference with observational data and unobserved confounding variables. *Ecology Letters*, 28(1):e70023.

Carretta, J. V., Oleson, E. M., Forney, K. A., Bradford, A. L., Yano, K., Weller, D. W., Lang, A. R., Baker, J., Orr, A. J., Hanson, B., Moore, J. E., Wallen, M., and Jr., R. L. B. (2024). U.S. Pacific marine mammal stock assessments: 2023. U.S. Department of Commerce, NOAA Technical Memorandum NMFS-SWFSC-704.

Carroll, C. and Miquelle, D. G. (2006). Spatial viability analysis of amur tiger panthera tigris altaica in the russian far east: the role of protected areas and landscape matrix in population persistence. *Journal of Applied Ecology*, 43(6):1056–1068.

Courbin, N., Besnard, A., Péron, C., Saraux, C., Fort, J., Perret, S., Tornos, J., and Grémillet, D. (2018). Short-term prey field lability constrains individual specialisation in resource selection and foraging site fidelity in a marine predator. *Ecology Letters*, 21(7):1043–1054.

Cove, M. V., Herrmann, V., Herrera, D. J., Augustine, B. C., Flockhart, D. T., and McShea, W. J. (2023). Counting the c apital's cats: Estimating drivers of abundance of free-roaming cats with a novel hierarchical model. *Ecological Applications*, 33(2):e2790.

Doty, M. S. and Oguri, M. (1956). The island mass effect. *ICES Journal of Marine Science*, 22(1):33–37.

Efford, M. G., Borchers, D. L., and Mowat, G. (2013). Varying effort in capture-recapture studies. *Methods in Ecology and Evolution*, 4(7):629–636.

Efford, M. G. and Schofield, M. R. (2022). A review of movement models in open population capture–recapture. *Methods in Ecology and Evolution*, 13(10):2106–2118.

Feloy, K., Powell, B. S., and Friedrich, T. (2024). Remote impacts of cyclonic eddies on productivity around the Main Hawaiian Islands. *Journal of Geophysical Research: Oceans*, 129(10):e2023JC020670.

Ferrão da Costa, G., Augustine, B. C., Fonseca, C., Mascarenhas, M., and Sutherland, C. (2026). Making the most of hard-won data: Incorporating commonly discarded information from unidentifiable samples improves precision and cost-effectiveness in wolf population monitoring. *Journal of Applied Ecology*, 63(6):e70468.

Fletcher, R. and Fortin, M. (2018). *Spatial ecology and conservation modeling*. Springer.

Friedrich, T., Powell, B., Gunnarson, J., Liu, G., Giardina, S., Stuecker, M., Hošeková, L., Feloy, K., and Stock, C. (2024). Submesoscale-permitting physical/biogeochemical future projections for the main hawaiian islands. *Journal of Advances in Modeling Earth Systems*, 16(2):e2023MS003855.

Friedrich, T., Powell, B., Stock, C., Hahn-Woernle, L., Dussin, R., and Curchitser, E. (2021). Drivers of phytoplankton blooms in Hawaii: A regional model study. *Journal of Geophysical Research: Oceans*, 126(5):e2020JC017069.

Gardner, B., McClintock, B. T., Converse, S. J., and Hostetter, N. J. (2022). Integrated animal movement and spatial capture-recapture models: Simulation, implementation, and inference. *Ecology*, 103(10):e3771.

Gardner, B., Reppucci, J., Lucherini, M., and Royle, J. A. (2010). Spatially explicit inference for open populations: estimating demographic parameters from camera-trap studies. *Ecology*, 91(11):3376–3383.

Gardner, B., Sollmann, R., Kumar, N. S., Jathanna, D., and Karanth, K. U. (2018). State space and movement specification in open population spatial capture-recapture models. *Ecology and evolution*, 8(20):10336–10344.

Henderson, E. E., Baird, R. W., Martin, S. W., Kratofil, M. A., Harnish, A. E., Mahaffy, S. D., Martin, C. R., and Southall, B. L. (2026). A review of the long-term study of resident odontocetes off Kaua'i and Ni'ihau and the analysis of exposures to mid-frequency active sonar. Technical report, The Effects of Noise on Aquatic Life IV.

Huggins, R. M. (1989). On the statistical analysis of capture experiments. *Biometrika*, 76(1):133–140.

Jacobson, E. K., Henderson, E. E., Miller, D. L., Oedekoven, C. S., Moretti, D. J., and Thomas, L. (2022). Quantifying the response of Blainville's beaked whales to US naval sonar exercises in Hawaii. *Marine Mammal Science*, 38(4):1549–1565.

Jolly, G. M. (1965). Explicit estimates from capture-recapture data with both death and immigration-stochastic model. *Biometrika*, 52(1/2):225–247.

Kareiva, P., Mullen, A., and Southwood, R. (1990). Population dynamics in spatially complex environments: theory and data. *Philosophical Transactions: Biological Sciences*, pages 175–190.

Kéry, M. and Schaub, M. (2011). *Bayesian population analysis using WinBUGS: a hierarchical perspective*. Academic press.

Kratofil, M. A., Harnish, A. E., Mahaffy, S. D., Henderson, E. E., Bradford, A. L., Martin, S. W., Lagerquist, B. A., Palacios, D. M., Oleson, E. M., and Baird, R. W. (2023). Biologically Important Areas II for cetaceans within US and adjacent waters–Hawai'i Region. *Frontiers in Marine Science*, 10:1053581.

Linden, D. W., Sirén, A. P., and Pekins, P. J. (2018). Integrating telemetry data into spatial capture-recapture modifies inferences on multi-scale resource selection. *Ecosphere*, 9(4):e02203.

McLoughlin, P. D., Morris, D. W., Fortin, D., Vander Wal, E., and Contasti, A. L. (2010). Considering ecological dynamics in resource selection functions. *Journal of Animal Ecology*, 79(1):4–12.

Merten, W., Reyer, A., Savitz, J., Amos, J., Woods, P., and Sullivan, B. (2016). Global fishing watch: Bringing transparency to global commercial fisheries. *arXiv preprint arXiv:1609.08756*.

Milleret, C., Dupont, P., Brøseth, H., Kindberg, J., Royle, J. A., and Bischof, R. (2018). Using partial aggregation in spatial capture recapture. *Methods in Ecology and Evolution*, 9(8):1896–1907.

Monnahan, C. C., Thorson, J. T., and Branch, T. A. (2017). Faster estimation of Bayesian models in ecology using Hamiltonian Monte Carlo. *Methods in Ecology and Evolution*, 8(3):339–348.

Nencioli, F., Kuwahara, V. S., Dickey, T. D., Rii, Y. M., and Bidigare, R. R. (2008). Physical dynamics and biological implications of a mesoscale eddy in the lee of Hawai'i: Cyclone Opal observations during E-Flux III. *Deep Sea Research Part II: Topical Studies in Oceanography*, 55(10-13):1252–1274.

Obermeyer, F., Bingham, E., Jankowiak, M., Pradhan, N., Chiu, J., Rush, A., and Goodman, N. (2019). Tensor variable elimination for plated factor graphs. In *International Conference on Machine Learning*, pages 4871–4880. PMLR.

Papaspiliopoulos, O., Roberts, G. O., and Sköld, M. (2007). A general framework for the parametrization of hierarchical models. *Statistical Science*, pages 59–73.

Partridge, D., Friedrich, T., and Powell, B. S. (2019). Reanalysis of the PacIOOS Hawaiian Island Ocean Forecast System, an implementation of the Regional Ocean Modeling System v3. 6. *Geoscientific Model Development*, 12(1):195–213.

Patchett, R., Smith, B. J., Thaxter, C. B., Burton, N. H., Franke, B. H., Yanco, S. W., Oliver, R. Y., Ellis-Soto, D., Tucker, M. A., Loveridge, A., et al. (2025). Habitat selection of three gull species in response to sudden changes in human mobility. *Proceedings of the Royal Society B: Biological Sciences*.

Phan, D., Pradhan, N., and Jankowiak, M. (2019). Composable effects for flexible and accelerated probabilistic programming in numpyro. *arXiv preprint arXiv:1912.11554*.

Rii, Y. M., Brown, S. L., Nencioli, F., Kuwahara, V., Dickey, T., Karl, D. M., and Bidigare, R. R. (2008). The transient oasis: Nutrient-phytoplankton dynamics and particle export in Hawaiian lee cyclones. *Deep Sea Research Part II: Topical Studies in Oceanography*, 55(10-13):1275–1290.

Rosel, P., Mullin, K., Garrison, L., Schwacke, L., Adams, J., Balmer, B., Conn, P., Conroy, M., Eguchi, T., Gorgone, A., Hohn, A., Mazzoil, M., Schwartz, C., Sinclair, C., Speakman, T., Urian, K., Vollmer, N., Wade, P., Wells, R., and Zolman, E. (2011). Photo-identification capture-mark-recapture techniques for estimating abundance of bay, sound and estuary populations of bottlenose dolphins along the U.S. East Coast and Gulf of Mexico: a workshop report.

Royle, J. A., Chandler, R. B., Sollmann, R., and Gardner, B. (2013a). *Spatial capture-recapture*. Academic press.

Royle, J. A., Chandler, R. B., Sun, C. C., and Fuller, A. K. (2013b). Integrating resource selection information with spatial capture-recapture. *Methods in Ecology and Evolution*, 4(6):520–530.

Royle, J. A. and Dorazio, R. M. (2008). *Hierarchical modeling and inference in ecology: the analysis of data from populations, metapopulations and communities*. Elsevier.

Royle, J. A., Dorazio, R. M., and Link, W. A. (2007). Analysis of multinomial models with unknown index using data augmentation. *Journal of Computational and Graphical Statistics*, 16(1):67–85.

Royle, J. A. and Young, K. V. (2008). A hierarchical model for spatial capture-recapture data. *Ecology*, 89(8):2281–2289.

Russell, R. E., Royle, J. A., Desimone, R., Schwartz, M. K., Edwards, V. L., Pilgrim, K. P., and Mckelvey, K. S. (2012). Estimating abundance of mountain lions from unstructured spatial sampling. *The Journal of Wildlife Management*, 76(8):1551–1561.

Seber, G. A. (1965). A note on the multiple-recapture census. *Biometrika*, 52(1/2):249–259.

Shaff, J. F. and Baird, R. W. (2021). Diel and lunar variation in diving behavior of rough-toothed dolphins (*Steno bredanensis*) off Kaua'i, Hawai'i. *Marine Mammal Science*, 37(4):1261–1276.

Smith, J. R. (2016). Multibeam backscatter and bathymetry synthesis for the main hawaiian islands final technical report july 2016.

Stock, C. A., Dunne, J. P., and John, J. G. (2014). Global-scale carbon and energy flows through the marine planktonic food web: An analysis with a coupled physical–biological model. *Progress in Oceanography*, 120:1–28.

Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., and Bürkner, P.-C. (2021). Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC (with discussion). *Bayesian analysis*, 16(2):667–718.

663 Yackulic, C. B., Dodrill, M., Dzul, M., Sanderlin, J. S., and Reid, J. A. (2020). A need for speed
664 in Bayesian population models: a practical guide to marginalizing and recovering discrete latent
665 states. *Ecological Applications*, 30(5):e02112.

A Covariate data

We investigated the effect of four covariates on space-use: bathymetric depth, a proxy of fishing effort, upward sea water velocity at 67 m (rough-toothed dolphin average dive depth; see below), and zooplankton concentration (Figures C.1, C.2, and C.3). For depth, we downloaded the Main Hawaiian Islands Multibeam Bathymetry and Backscatter Synthesis data, in 1-km regular grid, from the University of Hawai'i (Smith, 2016) (Figure C.1). We interpolated these data to a 4-km regular grid (see Section 2.2.1) with nearest neighbor interpolation. All covariates were standardized to have mean 0 and standard deviation of 1 (z-score standardization). There was no meaningful correlation between parameters (Figure C.4).

We gathered simulated zooplankton concentration estimates as well as vertical velocities from a physical/biogeochemical model of the Main Hawaiian Islands (Friedrich et al., 2021, 2024). The model is based on the Regional Ocean Modeling System coupled to the planktonic ecosystem model Carbon, Ocean Biogeochemistry and Lower Trophics Stock et al. (2014). The model was integrated for the time period 2010-2020 and embedded in a physical state-estimate using nearly 50 million observations (Partridge et al., 2019). Simulated depth-integrated zooplankton concentrations are shown by Figure C.2. Additionally, we suspected that upwelling could be an indicator of nutrient supply, and therefore prey availability in this oligotrophic system. We chose to use simulated vertical velocities at 67 m (Figure C.3) because it is the average dive depth for this population of rough-toothed dolphins (Shaff and Baird, 2021). We constructed a session-level covariate out of daily zooplankton concentration and upward sea water velocity data by averaging the covariate value for each date with effort—either days with photo-ID effort or days with telemetry fixes—in each session (Table B.1). For example, if a session had photo-ID effort on 02/07 and 02/08, and telemetry fixes on 02/07, 02/08, and 02/09, we averaged the zooplankton concentration across those three days.

We created a loose proxy for fishing effort, which we refer to as log AFE, by downloading Apparent Fishing Effort (AFE) from Global Fishing Watch data on a 0.01° regular grid(Merten et al., 2016). We summed the AFE data across this smaller grid to create a 4-km regular grid of total AFE, in hours, from 2012 to 2020 (Figure C.1). Note that the dataset began in 2012. We took the log of this value to get log AFE. We treated log AFE static covariate across years because it was too coarse to align with telemetry and photo-ID effort, as we did with zooplankton concentration and vertical velocity (see

above). The log AFE covariate is limited as a proxy for true fishing effort because it relies on automatic identification system (AIS) data. Only commercial fishing vessels longer than 65 feet (19.8 m) are required to carry AIS, and these represent a small subset of fishing vessels in our study area. Furthermore, we have no reason to believe fishing effort in itself a causal driver of rough-toothed dolphin space-use. Nevertheless, it seems reasonable that there might be some correlation between the preferences of these commercial vessels and productivity. For example, the abundance of rough-toothed dolphin prey sources, i.e., mesopelagic squid and fish (Shaff and Baird, 2021), might be correlated with commercial fish abundance, and which might be correlated with log AFE.

B Supplemental Tables

Table B.1: Description of sessions (primary periods) in SCR, where each year represents a session.

Year	Effort Start Date	Days with effort
2011	February 16	5
	July 21	18
2012	January 10	6
	June 12	21
2013	February 2	8
	July 26	7
2014	February 1	10
	October 7	11
2015	February 4	9
	September 3	9
2016	February 9	7
2017	August 4	10
2018	August 8	15
2019	-	0
2020	February 6	12

Table B.2: Summary of tagging effort, including the total duration of the deployment, the number of Douglas filtered locations (Kratofil et al., 2023), and the number of locations after thinning locations such that the remainder were at least 6 h apart.

Year	Type	ID	Duration	Total Count	After Thinning
2011	Spot5	SbTag001	7 d 14 h	125	37
	Spot5	SbTag002*	12 d 14 h	192	53
	Spot5	SbTag003*	18 d 13 h	280	73
2012	Spot5	SbTag004	27 d 18 h	392	123
	Mk10-A	SbTag005	6 d 3 h	81	6
	Mk10-A	SbTag006	10 d 11 h	145	25
	Mk10-A	SbTag007	13 d 17 h	224	64
2013	Mk10-A	SbTag008	3 d 18 h	64	15
	Mk10-A	SbTag009	9 d 22 h	154	20
	Mk10-A	SbTag010	13 d 10 h	214	17
2014	Mk10-A	SbTag011†	12 d 11 h	234	13
	Spot5	SbTag012	7 d 7 h	118	23
2015	Spot5	SbTag014	21 d 20 h	353	72
	Mk10-A	SbTag015	14 d 8 h	223	57
	Mk10-A	SbTag016	14 d 22 h	239	22
2016	Mk10-A	SbTag017	9 d 12 h	160	35
	Mk10-A	SbTag018	16 d 22 h	293	69
2017	Spot6	SbTag019	6 d 3 h	121	18
2018	Spot6	SbTag022	4 d 22 h	49	9
2021	Mk10-A	SbTag023‡	12 d 23 h	284	35
	Spot6	SbTag024‡	4 d 12 h	131	21
2022	SPLASH10	SbTag025‡	7 d 12 h	176	27

* These two may have been traveling together during the tag deployment.
† Omitted from the model for temporarily emigrating from the study area
‡ Omitted from the model for temporal mismatch

Table B.3: The total number of encounters, distinctive individuals, and non-distinctive individuals in each year. Note that counts only include individuals in good to excellent quality images.

Year	Encounters	Total Distinct	Total Count	Proportion Distinct
2011	15	159	191	0.86
2012	12	91	113	0.76
2013	4	27	33	0.77
2014	6	35	48	0.74
2015	11	226	270	0.83
2016	3	47	78	0.57
2017	8	112	152	0.76
2018	8	65	90	0.71
2020	6	30	46	0.75

Table B.4: Estimates of the total abundance N^{total} and the number of entrants B, Mean is the posterior mean, S.D. is the posterior standard deviation, C.V. is the standard deviation divided by the mean, and low and high are the lower and upper bounds, respectively, of the 90% credible interval.

Parameter	Year	Mean	S.D.	C.V.	Median	Low	High
N^{total}	2011	1997	232.6	0.12	1988	1628	2395
	2012	1855.0	202.7	0.11	1848	1532	2203
	2013	1759.1	187.7	0.11	1752	1460	2081
	2014	1680.0	178.3	0.11	1674	1396	1977
	2015	1644.9	172.8	0.11	1637	1375	1940
	2016	1616.9	188.3	0.12	1607	1325	1942
	2017	1546.9	204.1	0.13	1534	1234	1904
	2018	1464.3	220.4	0.15	1448	1130	1854
	2019	1439.0	259.6	0.18	1413	1060	1908
	2020	1405.2	293.4	0.21	1372	989	1940
B	2011	1652.4	184.9	0.11	1647	1360	1967
	2012	46.7	45.5	0.97	33	2	137
	2013	67.9	65	0.96	48	3	198
	2014	71.3	66.2	0.93	52	3	205
	2015	87.8	74.1	0.84	69	6	236
	2016	70.5	60.1	0.85	55	4	189
	2017	38.9	37.7	0.97	27	2	114
	2018	33.1	33	1.00	23	1	100
	2019	71.1	71.4	1.00	50	3	211
	2020	65.6	65.8	1.00	46	3	196

Table B.5: Parameter estimates from Model-16.

		Mean	SD	HDI 3%	HDI 97%	Mean MCSE	SD MCSE	ESS Bulk	ESS Tail	$\hat{R}$
μ_{α_0}		-7.473	0.459	-8.313	-6.584	0.015	0.008	962	1869	1.00
τ_{α_0}		1.28	0.306	0.766	1.861	0.007	0.005	1966	2862	1.00
α_0	2011	-6.317	0.194	-6.676	-5.951	0.003	0.002	3939	4792	1.00
α_0	2012	-7.989	0.223	-8.425	-7.581	0.004	0.002	2982	3993	1.00
α_0	2013	-8.71	0.338	-9.333	-8.057	0.005	0.004	4533	4347	1.00
α_0	2014	-9.49	0.277	-10.003	-8.963	0.005	0.003	3557	5012	1.00
α_0	2015	-7.844	0.229	-8.271	-7.411	0.004	0.002	2761	4671	1.00
α_0	2016	-5.976	0.25	-6.451	-5.511	0.004	0.002	3939	5223	1.00
α_0	2017	-6.164	0.228	-6.604	-5.758	0.005	0.002	2339	4133	1.00
α_0	2018	-7.983	0.296	-8.544	-7.438	0.006	0.003	2395	4258	1.00
α_0	2019	-7.457	1.414	-10.086	-4.685	0.023	0.02	3588	4086	1.00
α_0	2020	-7.889	0.413	-8.684	-7.142	0.009	0.005	2170	3689	1.00
σ	2011	15.534	1.114	13.542	17.671	0.038	0.02	887	1555	1.00
σ	2012	30.325	1.87	26.94	33.905	0.051	0.031	1366	1915	1.00
σ	2013	21.891	1.842	18.39	25.223	0.04	0.024	2088	3013	1.00
σ	2014	40.096	4.837	31.598	49.072	0.104	0.078	2281	2619	1.00
σ	2015	31.944	2.307	27.831	36.281	0.085	0.05	763	1107	1.00
σ	2016	16.216	1.013	14.398	18.19	0.019	0.013	2890	3426	1.00
σ	2017	24.282	2.449	19.826	28.977	0.074	0.04	1094	2269	1.00
σ	2018	26.919	3.033	21.405	32.62	0.086	0.046	1227	2134	1.00
σ	2019	20.965	6.213	10.252	32.335	0.082	0.09	5856	4651	1.00
σ	2020	30.565	5.22	21.032	40.078	0.143	0.095	1329	2545	1.00
α_2		1.741	0.122	1.506	1.963	0.003	0.001	2291	4034	1.00
α_3		-1.11	0.064	-1.232	-0.994	0.001	0.001	2884	4058	1.00
α_4		0.323	0.022	0.282	0.366	0	0	4455	4304	1.00
μ_{α_5}		0.073	0.059	-0.041	0.184	0.001	0.001	1730	2394	1.00
τ_{α_5}		0.146	0.07	0.028	0.281	0.002	0.002	1248	712	1.00
α_5	2011	0.147	0.03	0.09	0.206	0.001	0	3387	4332	1.00
α_5	2012	0.128	0.057	0.027	0.241	0.001	0.001	7956	5502	1.00
α_5	2013	0.041	0.062	-0.076	0.153	0.001	0.001	6080	5618	1.00
α_5	2014	0.149	0.09	-0.02	0.32	0.001	0.001	4676	5097	1.00
α_5	2015	0.093	0.048	0.004	0.184	0.001	0.001	8444	5057	1.00

α_5	2016	0.12	0.061	0.007	0.24	0.001	0.001	6653	4693	1.00
α_5	2017	0.169	0.078	0.03	0.322	0.001	0.001	4923	4109	1.00
α_5	2018	-0.129	0.111	-0.333	0.075	0.003	0.002	1759	799	1.00
α_5	2019	0.073	0.171	-0.26	0.382	0.002	0.003	5143	3901	1.00
α_5	2020	-0.068	0.094	-0.242	0.105	0.002	0.001	2275	1218	1.00
μ_{α_6}		0.264	0.422	-0.545	1.051	0.013	0.009	1032	1895	1.00
τ_{α_6}		1.196	0.346	0.57	1.808	0.007	0.005	2192	3177	1.00
α_6	2011	0.427	0.134	0.181	0.681	0.002	0.001	5319	5209	1.00
α_6	2012	0.137	0.149	-0.149	0.412	0.002	0.002	7237	5345	1.00
α_6	2013	1.346	0.399	0.619	2.096	0.005	0.004	5743	5260	1.00
α_6	2014	0.704	0.357	0.03	1.37	0.004	0.004	7861	4543	1.00
α_6	2015	1.211	0.163	0.915	1.52	0.002	0.002	6506	5414	1.00
α_6	2016	-1.108	0.292	-1.686	-0.584	0.004	0.003	4385	4243	1.00
α_6	2017	0.991	0.349	0.333	1.628	0.005	0.004	5466	5163	1.00
α_6	2018	0.795	0.236	0.362	1.249	0.003	0.003	6337	5284	1.00
α_6	2019	0.247	1.297	-2.293	2.64	0.018	0.02	5153	3976	1.00
α_6	2020	-1.959	0.87	-3.552	-0.333	0.013	0.011	4447	3653	1.00
ψ		0.369	0.034	0.309	0.437	0.001	0	1708	3132	1.00
ϕ		0.896	0.019	0.86	0.933	0	0	2384	2940	1.00
π	2011	0.741	0.064	0.622	0.857	0.001	0.001	2386	3438	1.00
π	2012	0.022	0.021	0	0.06	0	0	5193	3108	1.00
π	2013	0.032	0.031	0	0.088	0	0.001	3914	2203	1.00
π	2014	0.035	0.032	0	0.093	0	0	4027	2549	1.00
π	2015	0.044	0.036	0	0.111	0	0	4533	3206	1.00
π	2016	0.034	0.028	0	0.084	0	0	4404	2848	1.00
π	2017	0.017	0.016	0	0.047	0	0	4998	3488	1.00
π	2018	0.015	0.014	0	0.04	0	0	6481	3959	1.00
π	2019	0.031	0.03	0	0.085	0	0.001	4631	4188	1.00
π	2020	0.029	0.027	0	0.079	0	0	5519	4588	1.00

C Supplemental Figures

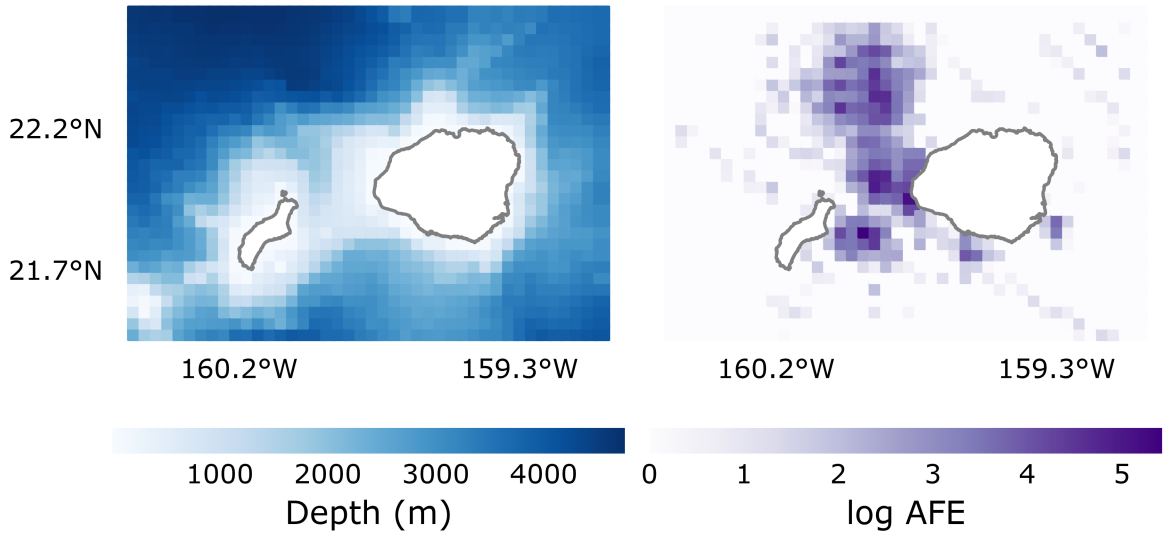


Figure C.1: The depth (blue cells) and log fishing effort (purple cells) for every grid cell in the state space.

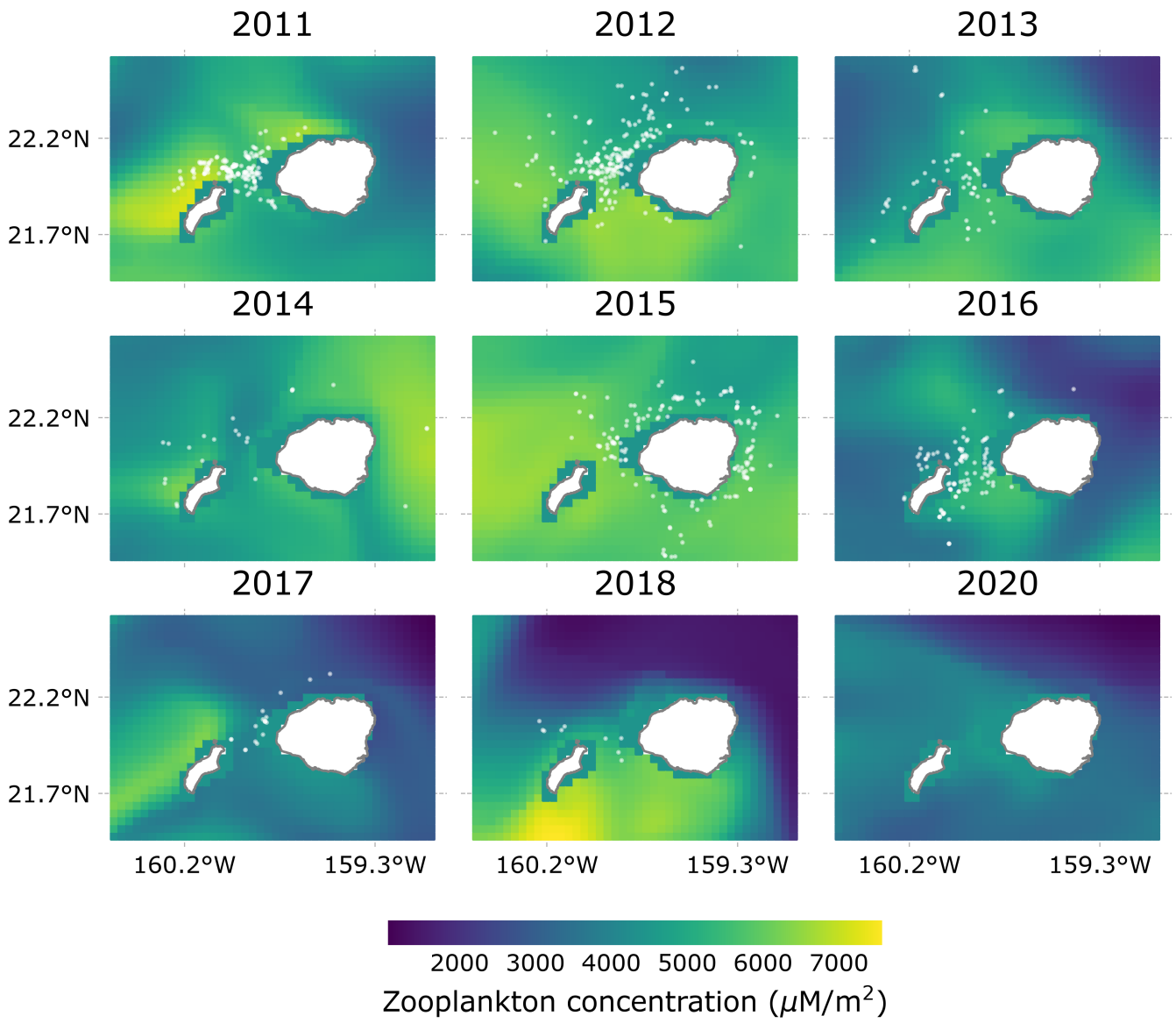


Figure C.2: Vertically integrated concentration of zooplankton (colored cells) in every grid cell in the state space. Telemetry locations (white dots) were thinned such that there was only one location per 6 h window.

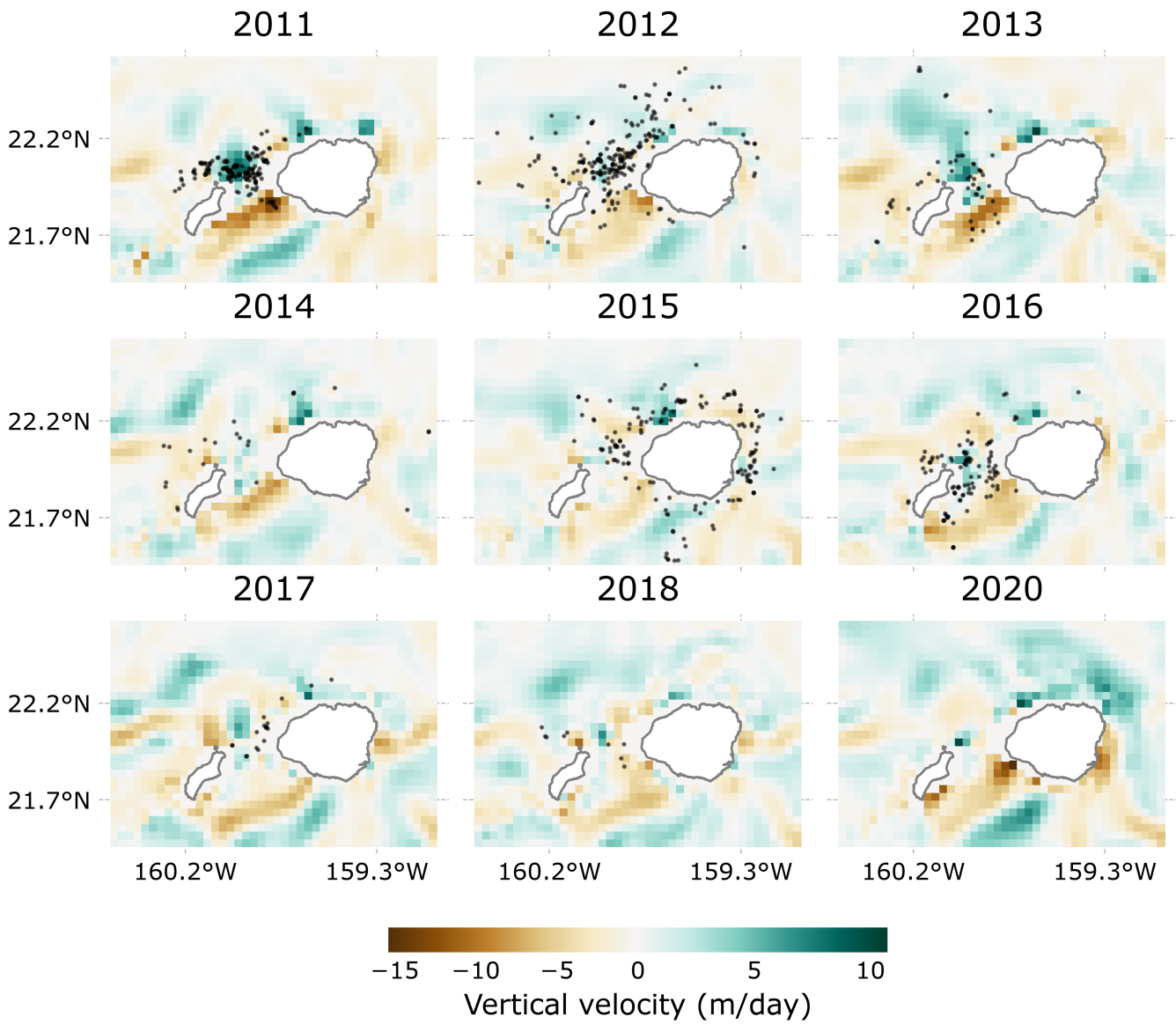


Figure C.3: Upward sea water velocity (colored cells) for each grid cell in the state space. Telemetry locations (black dots) were thinned such that there was only one location per 6 h window.

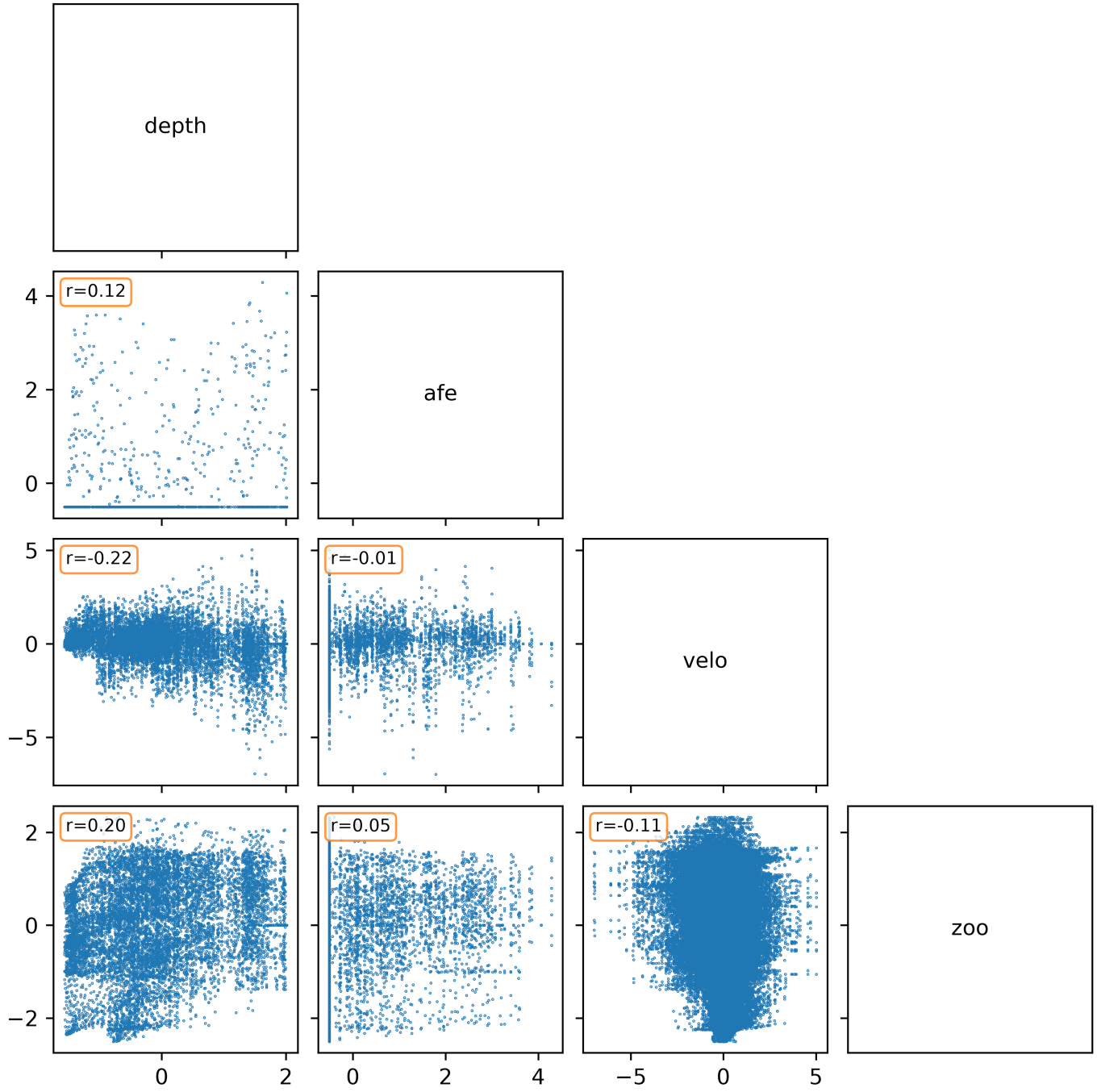


Figure C.4: Value of pairs of covariates (blue dot) at every grid cell, as well as the the Pearson's correlation coefficient between pairs of covariates (upper left text).

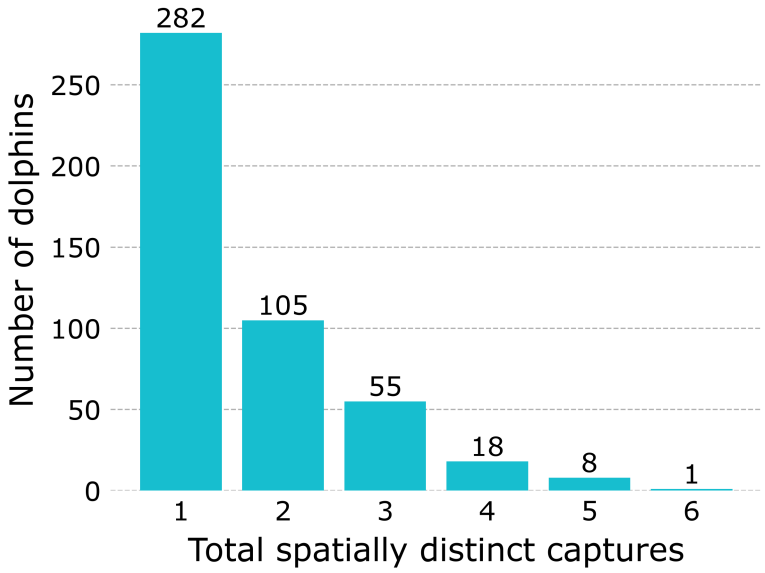


Figure C.5: The number of individuals (blue bar) who were captured in x distinct locations.

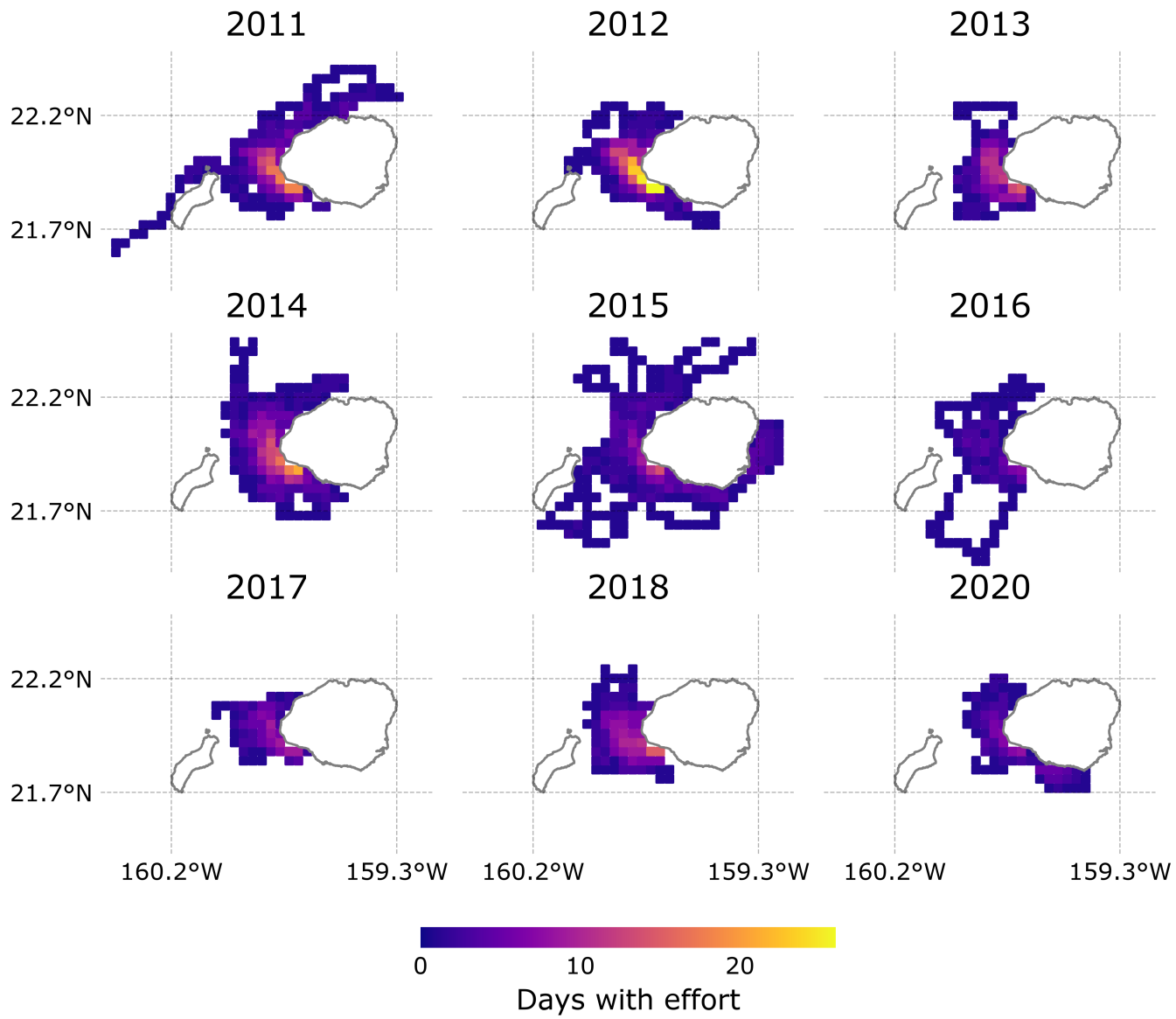


Figure C.6: The number of survey days (colored cells) that each grid cell was visited. The eastings and northings are represented by x and y, respectively.

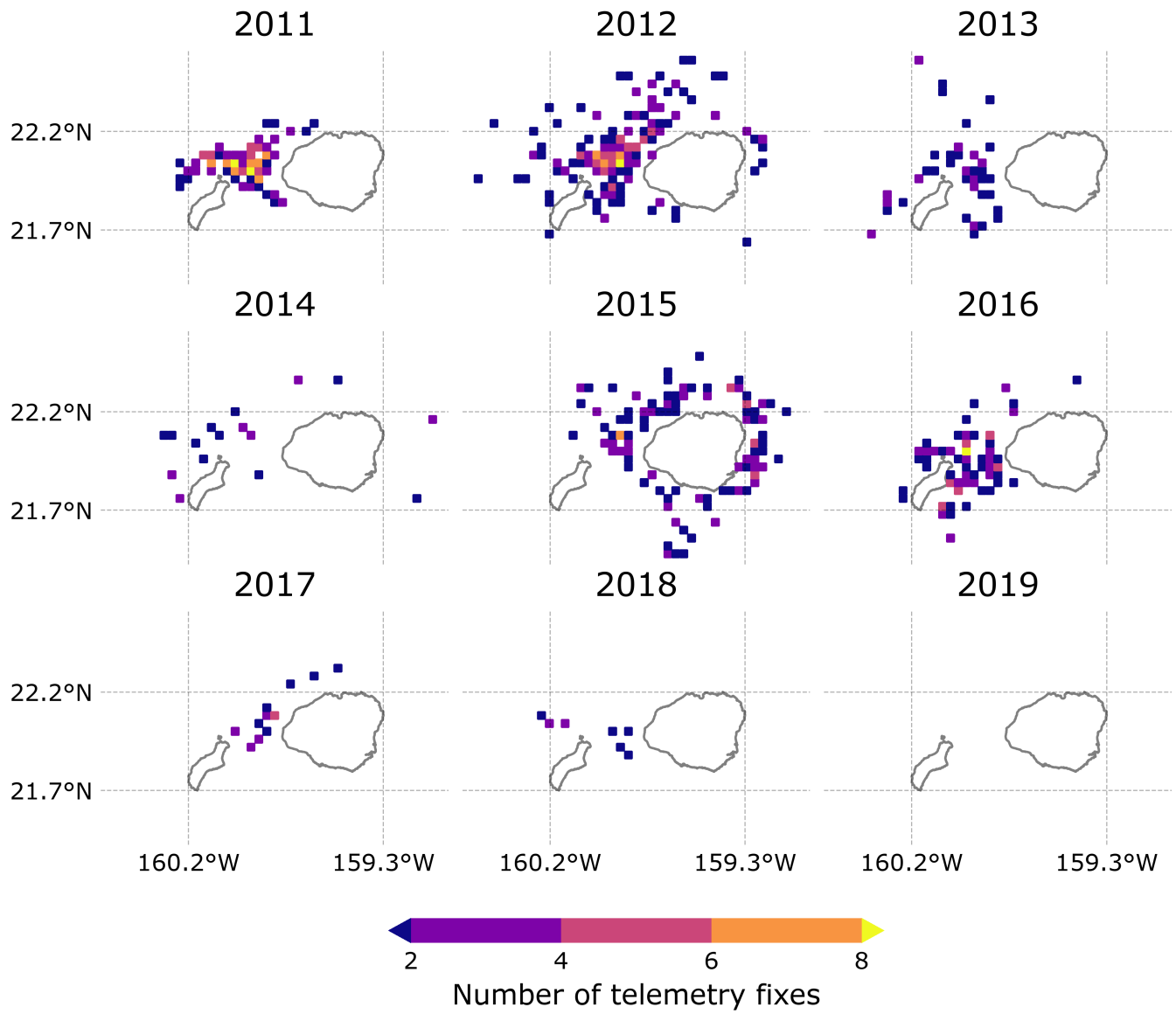


Figure C.7: The number of telemetry fixes (colored squares), after thinning the data such that there was only one fix per 6 h window per individual, in each grid cell for all tagged individuals in each year.

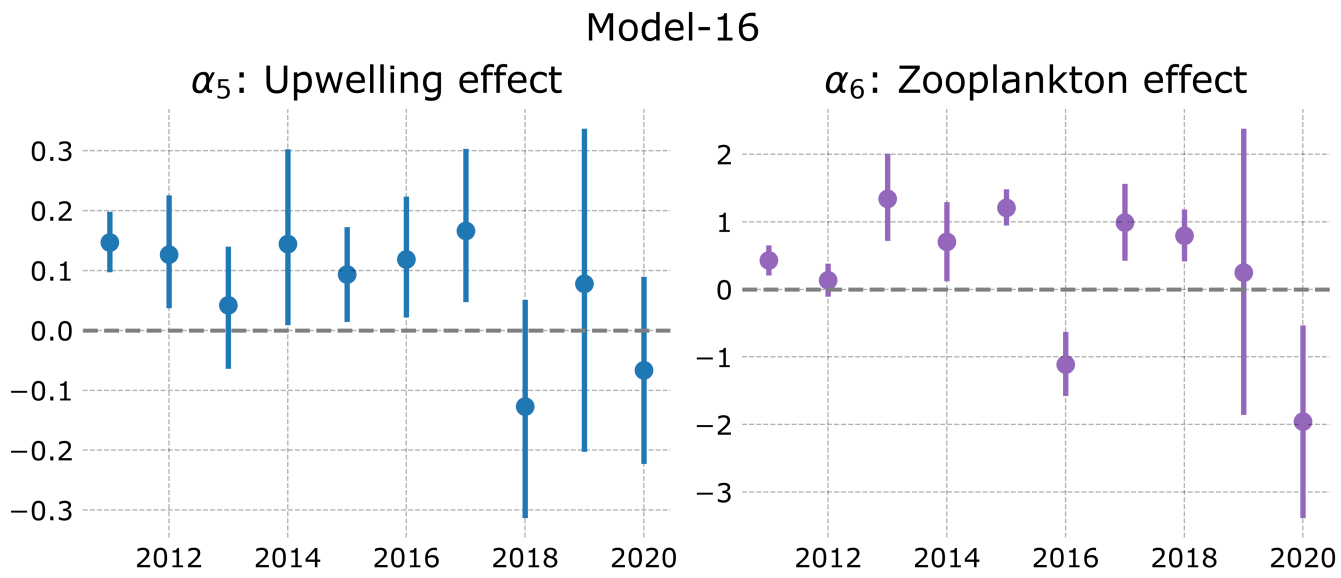


Figure C.8: The posterior median (circles) and 90% credible interval (vertical line) for the estimates of α_5, the effect of upwelling, i.e., vertical velocity, on use and α_6, the effect of zooplankton concentration on use. The estimates are from model-16, which included time-varying effects for both covariates.

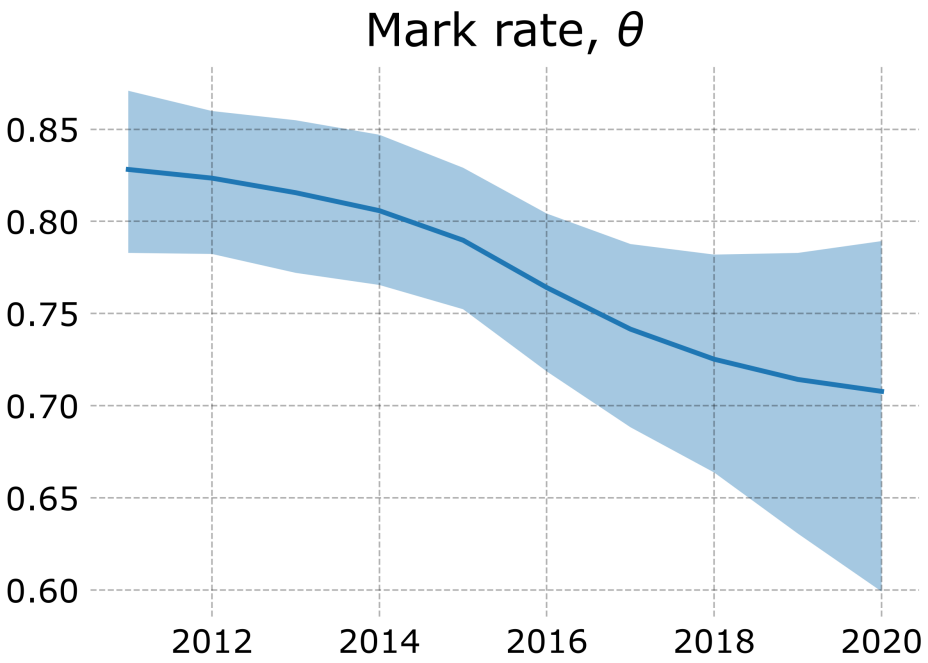


Figure C.9: The predicted mark rate (blue line) and 90% credible intervals (shaded area) over time.

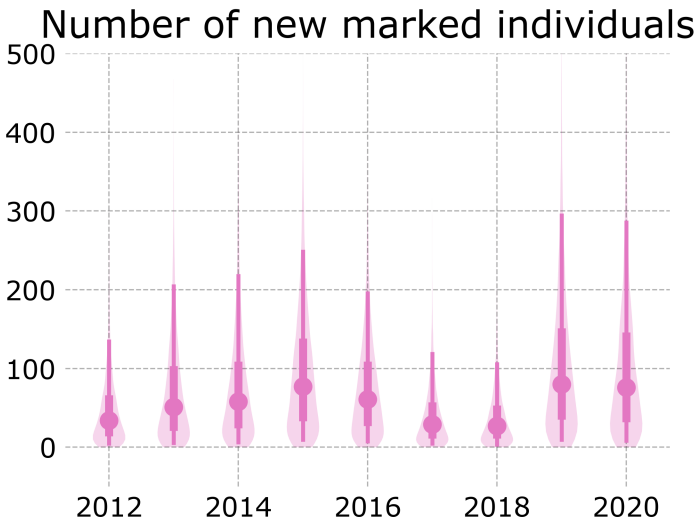


Figure C.10: Posterior distribution (violin), 50% credible interval (thick pink line), 90% credible interval (thin pink line), and posterior median (pink dot) of the number of entrants, i.e., new marked individuals, into the population each year.

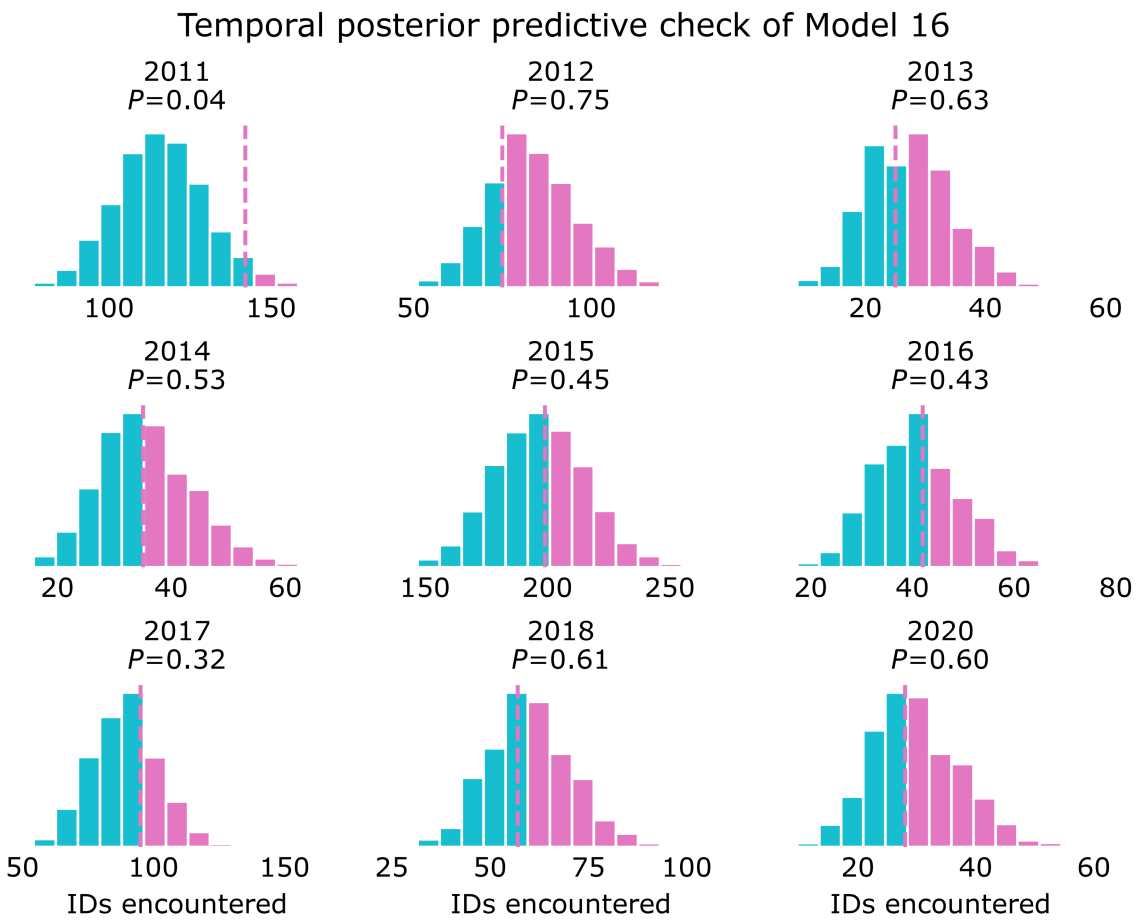


Figure C.11: Posterior predictive samples (colored bars) and observed value (pink dashed line) of the number of unique individuals encountered during each year. Bins of samples that exceeded the observed value are colored pink. The proportion of samples that exceeded the observed value is the Bayesian *P*-value.

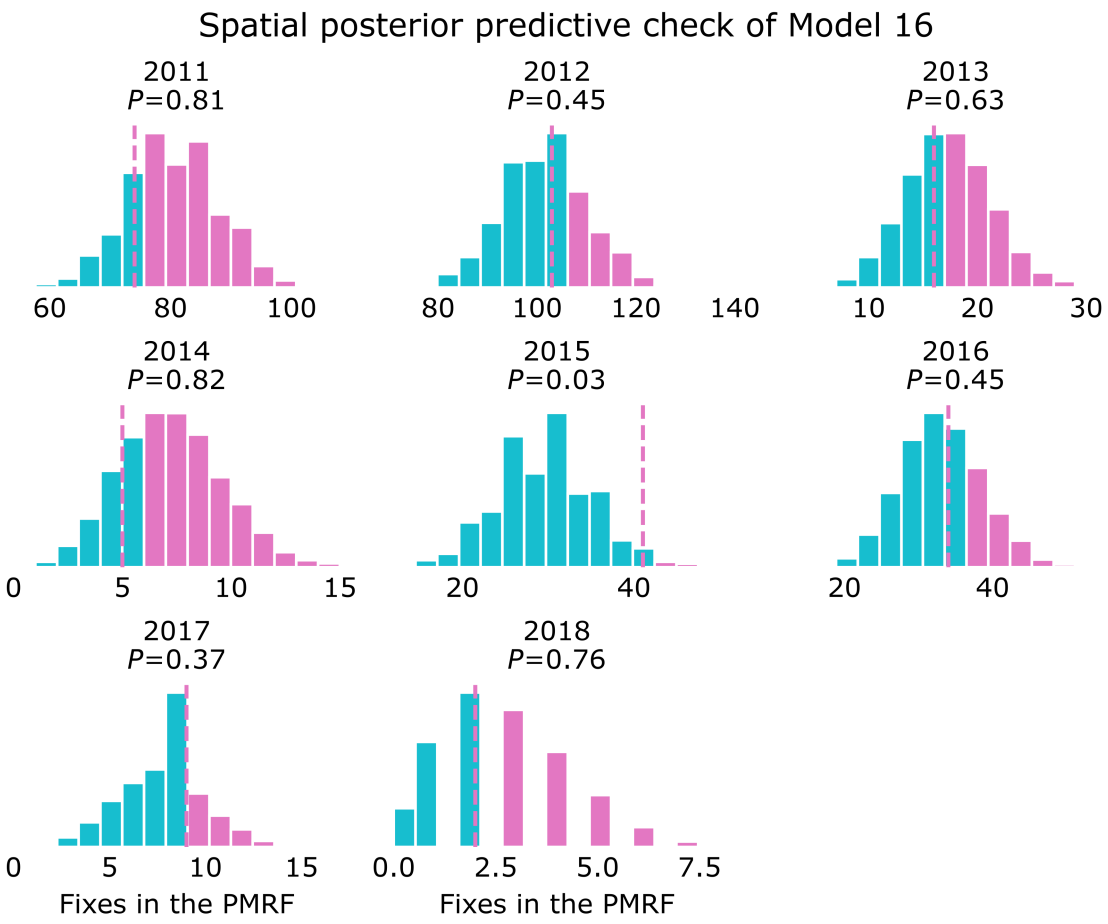


Figure C.12: Posterior predictive samples (colored bars) and observed value (pink dashed line) of the number of telemetry fixes within the US Navy's Pacific Missile Range Facility during each year. Bins of samples that exceeded the observed value are colored pink. The proportion of samples that exceeded the observed value is the Bayesian *P*-value.

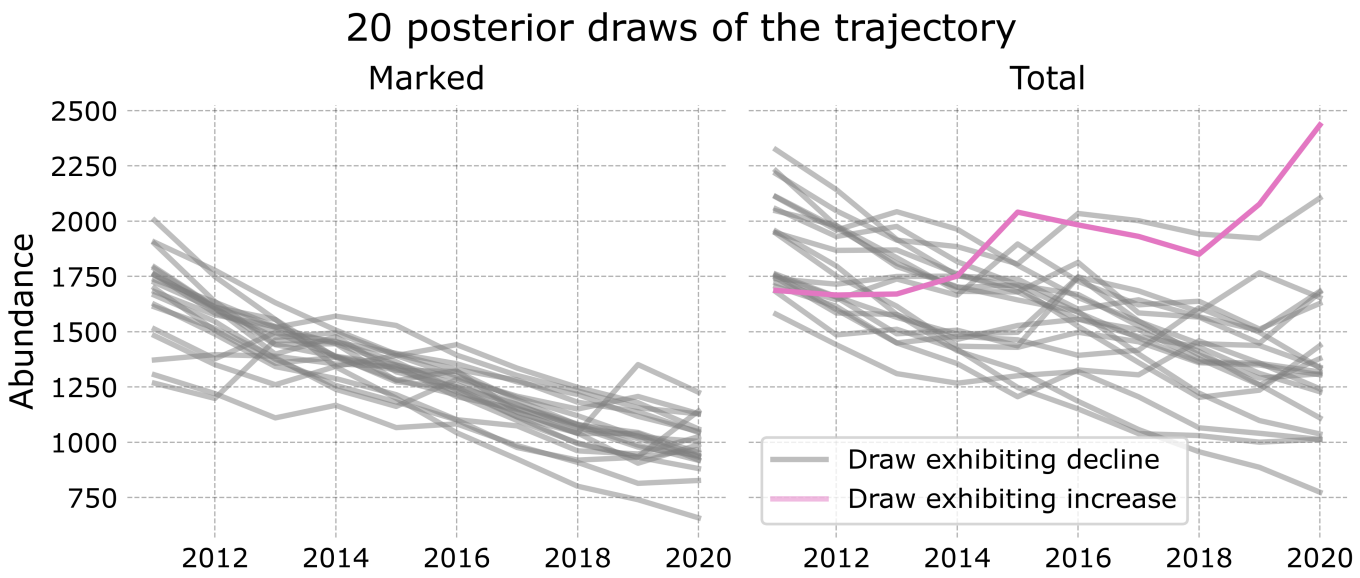


Figure C.13: Twenty random posterior draws of the population trajectory (lines), where pink lines indicate of that the trajectory declined and gray lines indicate otherwise.

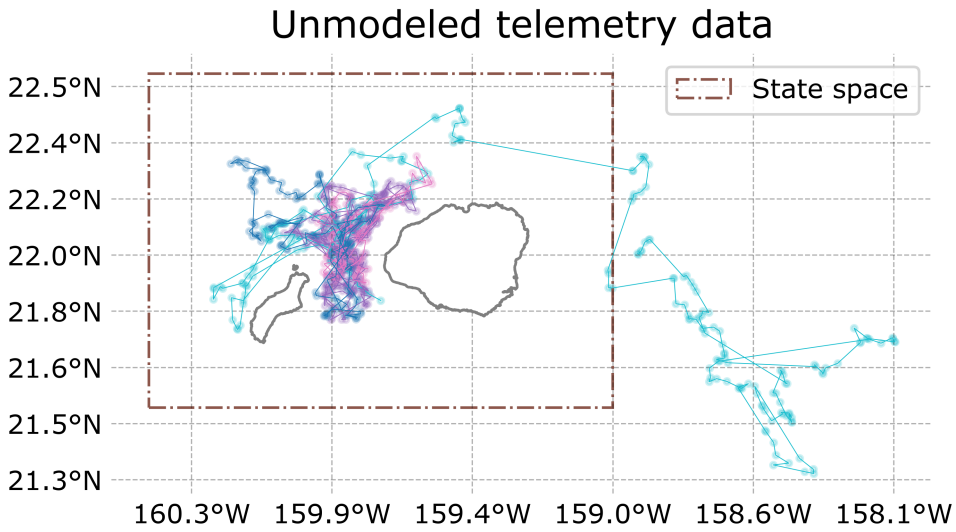


Figure C.14: Telemetry locations for four tagged dolphins (colored dots and lines) that were omitted from the analysis. One individual (cyan dots and lines) was omitted for temporarily exiting the state-space, while the others were omitted for lack of temporal alignment (Table B.2).

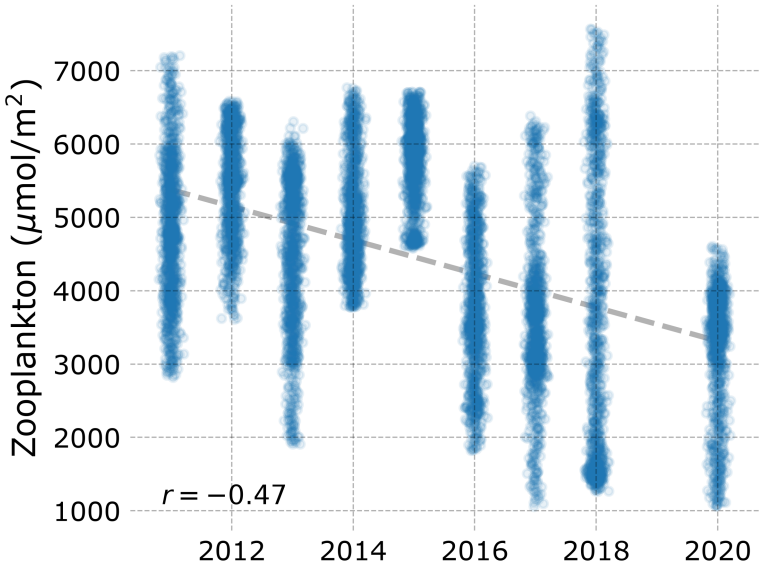


Figure C.15: Vertically integrated zooplankton concentration in each grid cell (blue dots) over time. The Pearson's correlation coefficient (bottom left text) and simple linear regression fit (gray dashed line) show the relationship between time and zooplankton concentration.